

72nd BPSC PRELIMS

प्रहार

Test Series

Test- 28/12

FULL LENGTH TEST

Solution

English



BPSC TEST-28/12 (SOLUTION)

Q 1. Ans. C

Explanation:

- **Statement 1 is correct:** a sponge is a solid because it has a definite shape and a rigid framework. Although it can be compressed, this is due to the presence of **tiny air-filled pores**. When pressure is applied, the air escapes from these pores, making the sponge appear compressible. The solid framework itself remains intact; hence, a sponge is classified as a solid.
- **Statement 2 is incorrect:** Ice floats on water because it has a **lower density**, not a higher density, than liquid water. In ice, water molecules are arranged in an **open hexagonal hydrogen-bonded lattice**, which increases the volume occupied by the same mass of water. Since density is mass divided by volume, this larger volume results in a **lower density** than liquid water. Consequently, ice floats on water.
- **Statement 3 is correct:** solids have **definite shape, fixed volume, well-defined boundaries**, and are **almost incompressible** because their constituent particles are closely packed with very little intermolecular space. Strong intermolecular forces keep the particles fixed in position, allowing only vibrational motion.

Q 2. Ans. A

Explanation:

- **Statement 1 is correct:** one of the fundamental postulates of **Bohr's atomic model** is that electrons revolve around the nucleus only in certain **fixed circular orbits (stationary states)**. Each orbit has a definite energy, and electrons moving in these permitted orbits **do not radiate energy**. Thus, the statement is correct.
- **Statement 2 is incorrect:** In Bohr's model, an electron **absorbs energy** to move from a lower energy orbit to a higher energy orbit (excitation). It **emits energy only when it falls from a higher energy orbit to a lower energy orbit**, releasing the energy difference as electromagnetic radiation.
- **Statement 3 is incorrect:** Bohr's atomic model successfully explains the spectrum of **hydrogen and hydrogen-like (single-electron) atoms** such as He^+ and Li^{2+} . However, it **fails to explain the spectra of multi-electron atoms**, as well as phenomena such as fine spectral lines, the Zeeman effect, and the Stark effect. Therefore, the statement is incorrect.

Q 3. Ans. B

Explanation:

- **Pair 1 is incorrect:** the s-block consists of elements whose valence electron enters the **s-orbital**. It includes **Group 1 (alkali metals)** and **Group 2 (alkaline earth metals)**, along with hydrogen and helium (based on electronic configuration). **Transition metals belong to the d-block**, not the s-block. Therefore, this pair is incorrectly matched.
- **Pair 2 is correct:** The p-block comprises **Groups 13 to 18** of the periodic table. These elements are commonly referred to as **main group (representative) elements** and include metals, metalloids, non-metals, halogens, and noble gases. Therefore, this pair is correctly matched.
- **Pair 3 is incorrect:** The d-block consists of **transition elements** (Groups 3 to 12), whose differentiating electron enters the **d-orbital**. **Alkali metals** (Group 1) and **alkaline earth metals** (Group 2) belong to the s-block, not the d-block. Hence, this pair is incorrect.
- **Pair 4 is correct:** The f-block includes the **lanthanoids and actinoids**, collectively known as the **inner transition elements**. In these elements, the differentiating electron enters the **f-orbital**. Therefore, this pair is correctly matched.

Q 4. Ans. C

Explanation:

- **Calcination** is the process of **heating carbonate or hydrated ores in the absence or limited supply of air**. This process removes moisture, volatile impurities, and converts carbonate ores into their corresponding oxides.
- **Roasting** involves **heating sulphide ores strongly in excess air or oxygen**. During roasting, sulphide ores are converted into metal oxides with the evolution of sulphur dioxide gas.
- **Smelting** is the process of **reducing metal oxides to the metallic state** by heating them with a suitable reducing agent, such as **carbon or carbon monoxide**, in a furnace. It is a major step in the extraction of metals from their ores.
- **Leaching** is a concentration process in which the **desired ore is selectively dissolved in a suitable chemical reagent**, leaving behind insoluble impurities. The dissolved metal is later recovered from the solution.

Q 5. Ans. B

Explanation:

- Solutions are classified based on the physical states of the **solute** and the **solvent**. A **gas in a solid solution** is one in which the **solute is a gas** and the **solvent is a solid**.
- **Hydrogen in palladium** is a classic example of a **gas dissolved in a solid**. Palladium has the remarkable ability to **absorb large volumes of hydrogen gas** into its crystal lattice without undergoing any significant chemical change. This property makes palladium useful in hydrogen purification and storage technologies. Therefore, hydrogen in palladium is an example of a **gas in a solid solution**.

Q 6. Ans. C

Explanation:

- **Statement 1 is correct:** Colloidal particles are **small enough to pass through ordinary filter paper** but **cannot pass through semipermeable membranes**. This property distinguishes colloids from suspensions, whose particles are retained by ordinary filter paper. Therefore, this statement is **correct**.
- **Statement 2 is correct:** **Tyndall effect** is the phenomenon in which **colloidal particles scatter the path of a beam of light**, making it visible. The effect occurs because colloidal particles are large enough to scatter light but small enough to remain dispersed in the medium. Therefore, this statement is **correct**.
- **Statement 3 is incorrect:** A suspension exhibits the Tyndall effect **only while its particles remain suspended** in the dispersion medium. When left undisturbed, the particles **settle at the bottom due to gravity**, leaving the upper liquid nearly clear. Once the particles have settled, they are no longer available to scatter light effectively, and the **Tyndall effect disappears**. Therefore, the statement is **incorrect**.

Q 7. Ans. B

Explanation:

- **Statement 1 is incorrect:** **thermosetting plastics do not soften on heating**. During their manufacture, they undergo irreversible chemical changes that produce a rigid, three-dimensional network. Once set, they cannot be softened by reheating.
- **Statement 2 is correct:** Thermosetting plastics undergo **extensive cross-linking** between polymer chains during heating. This cross-linking forms a rigid three-dimensional structure, making them

hard, heat-resistant, and permanently set. Common examples include **Bakelite** and **melamine**.

- **Statement 3 is incorrect:** Thermosetting plastics **cannot be melted and remoulded** after they have been set. On strong heating, they **decompose or char instead of melting** because of their highly cross-linked structure. Therefore, they cannot be recycled by melting like thermoplastics.

Q 8. Ans. A

Explanation:

- **Statement 1 is correct:** Graphene is a **two-dimensional allotrope of carbon** consisting of a **single layer of carbon atoms** arranged in a **hexagonal (honeycomb) lattice**. It is the basic structural unit of graphite and is one atom thick. Owing to its exceptional strength, high electrical conductivity, thermal conductivity, and flexibility, graphene has numerous applications in electronics, energy storage, sensors, and advanced composite materials.
- **Statement 2 is incorrect:** through **sp^2 hybridisation**, forming three strong **σ (sigma) bonds**. The remaining unhybridized **p-orbital** contains one electron that participates in a **delocalized π -electron system**, which is responsible for graphene's excellent electrical conductivity. **sp^3 hybridisation**, where each carbon forms four single bonds, is characteristic of **diamond**, not graphene.

Q 9. Ans. D

Explanation:

- **Statement:** **Ribosomes are present in both prokaryotic and eukaryotic cells**. They are the sites of **protein synthesis** and occur either freely in the cytoplasm or attached to the rough endoplasmic reticulum in eukaryotic cells. Prokaryotes possess **70S ribosomes**, whereas eukaryotic cytoplasm contains **80S ribosomes** (with 70S ribosomes present in mitochondria and chloroplasts).
- **Conclusion I:** The presence of ribosomes in both prokaryotic and eukaryotic cells indicates that **protein synthesis is an essential and universal process** required for growth, repair, metabolism, and other cellular functions. Therefore, this conclusion is correct.
- **Conclusion II:** Ribosomes are **non-membrane-bound organelles**. They are composed of **ribosomal RNA (rRNA) and proteins** and lack a surrounding membrane. This is true for both prokaryotic and eukaryotic cells. Hence, the conclusion is incorrect.

Q 10. Ans. C

Explanation:

- **Statement:** mitochondria and chloroplasts are **semiautonomous organelles** because they contain their own **circular DNA, RNA, and 70S ribosomes**. Although they depend on the nucleus for many proteins, they retain the ability to express some of their own genes.
- **Conclusion I:** Since mitochondria and chloroplasts possess **their own DNA and ribosomes**, they can **transcribe and translate some of their own genes**, enabling them to synthesise a portion of the proteins required for their functioning. However, many other proteins are encoded by nuclear genes and imported into these organelles.
- **Conclusion II:** mitochondria and chloroplasts are **self-replicating organelles**. New mitochondria and chloroplasts are formed **only by the growth and division of pre-existing mitochondria and chloroplasts**; they are **not synthesised de novo** by the cell. This property supports the **endosymbiotic theory** regarding their origin.

Q 11. Ans. A

Explanation:

- **Statement:** The **stratospheric ozone layer** acts as a protective shield by absorbing most of the Sun's **harmful ultraviolet (UV-B) radiation**. This prevents excessive UV radiation from reaching the Earth's surface, thereby protecting living organisms from DNA damage, skin cancer, cataracts, and other harmful effects.
- **Conclusion I:** This conclusion is incorrect. Although **ozone is a greenhouse gas**, it is **not the primary greenhouse gas responsible for global warming**. The major contributors to the enhanced greenhouse effect are **carbon dioxide (CO₂), methane (CH₄), nitrous oxide (N₂O), and chlorofluorocarbons (CFCs)**. Therefore, ozone is not the principal cause of global warming. This conclusion is incorrect.
- **Conclusion II:** **Stratospheric ozone** is beneficial because it absorbs harmful ultraviolet radiation. However, **tropospheric (ground-level) ozone** is a **harmful air pollutant** formed through photochemical reactions involving nitrogen oxides and volatile organic compounds. It can damage lung tissues, reduce crop yields, and harm vegetation. Therefore, increasing ozone at **every altitude** is not beneficial. This conclusion is also incorrect.

Q 12. Ans. C

Explanation:

- The **mitral valve**, also known as the **bicuspid valve**, is located between the **left atrium** and the **left ventricle**. It consists of **two cusps (flaps)** and ensures the **one-way flow of blood** from the left atrium to the left ventricle.
- During **ventricular contraction (ventricular systole)**, the pressure inside the left ventricle rises sharply. The **mitral valve closes**, preventing the backflow of oxygenated blood into the left atrium. This maintains efficient circulation and ensures that blood is pumped into the **aorta** through the aortic valve.

Q 13. Ans. C

Explanation:

- The **hypothalamus** is a part of the **forebrain (diencephalon)** that acts as the **main centre for maintaining homeostasis** in the body. One of its most important functions is the **regulation of body temperature (thermoregulation)**. It continuously monitors the temperature of the blood and initiates appropriate responses to maintain the body's normal temperature. For example, when body temperature rises, the hypothalamus stimulates **sweating and dilation of skin blood vessels** to promote heat loss. Conversely, when body temperature falls, it triggers **shivering and constriction of blood vessels** to conserve and generate heat.

Q 14. Ans. B

Explanation:

- During **strenuous exercise**, the rate of cellular respiration increases to meet the higher energy demand of muscles. As a result, **more carbon dioxide (CO₂)** is produced and released into the blood.
- The increased concentration of **CO₂** (and the associated rise in hydrogen ion concentration, causing a fall in blood pH) is detected by **chemoreceptors** located in the **carotid bodies, aortic bodies, and medulla oblongata**. These receptors stimulate the respiratory centre in the medulla, which increases both the **rate and depth of breathing**. This enhanced ventilation helps remove excess CO₂ and maintain normal blood pH.

Q 15. Ans. A

Explanation:

- **Bile** is secreted by the **liver** and stored in the **gall bladder**. It contains **bile salts, bile pigments,**

cholesterol, phospholipids, and bicarbonate ions, but does not contain digestive enzymes.

- The **primary function of bile salts** is to **emulsify fats**. During emulsification, large fat globules are broken down into numerous tiny droplets, thereby greatly increasing the surface area available for the action of the enzyme **pancreatic lipase**. This process makes the digestion and subsequent absorption of fats much more efficient. Bile salts also aid in the absorption of **fat-soluble vitamins (A, D, E, and K)**.

Q 16. Ans. C

Explanation:

- **Malaria** is caused by the **protozoan parasite Plasmodium**, mainly **Plasmodium vivax**, **P. falciparum**, **P. malariae**, and **P. ovale**. The disease is transmitted to humans through the bite of an **infected female Anopheles mosquito**, which acts as the vector. After entering the human body, the parasite first multiplies in the **liver cells** and then infects **red blood cells (RBCs)**, causing periodic fever, chills, sweating, and anaemia.
- **Typhoid** is caused by the **bacterium Salmonella typhi**.
- **Ringworm** is a **fungal disease** caused by species of **Microsporum**, **Trichophyton**, or **Epidermophyton**.
- **Tuberculosis (TB)** is caused by the **bacterium Mycobacterium tuberculosis**.

Q 17. Ans. C

Explanation:

- **Typhoid fever**, caused by the bacterium **Salmonella typhi**, is commonly diagnosed using the **Widal test**. This test detects **specific antibodies (agglutinins)** produced in the patient's serum against the **O (somatic) and H (flagellar) antigens** of **Salmonella typhi**. A significant antibody titre, along with the patient's clinical symptoms, supports the diagnosis of typhoid fever.
- **ELISA (Enzyme-Linked Immunosorbent Assay)**: Used to detect specific **antigens or antibodies** in various diseases, including **HIV infection**, hepatitis, and several other infectious and autoimmune disorders. It is **not the standard test for typhoid**.
- **Mantoux test**: A **tuberculin skin test** used to detect exposure to **Mycobacterium tuberculosis** (tuberculosis).
- **Western blot**: Primarily used as a **confirmatory test for HIV infection** by detecting antibodies against HIV proteins.

Q 18. Ans. C

Explanation:

- **Fish oils and most vegetable oils are rich in unsaturated fatty acids**. These fatty acids contain **one or more double bonds (C=C)** between carbon atoms, making them **liquid at room temperature**.
- **Unsaturated fatty acids are of two types**:
 - ✓ **Monounsaturated fatty acids (MUFAs)**: Contain one double bond (e.g., oleic acid).
 - ✓ **Polyunsaturated fatty acids (PUFAs)**: Contain two or more double bonds, including **omega-3 fatty acids** found abundantly in fish oils.
- These fats are generally considered **heart-healthy** because they help reduce low-density lipoprotein (LDL) cholesterol and may lower the risk of cardiovascular diseases.

Q 19. Ans. B

Explanation:

- According to **Newton's Second Law of Motion**, the acceleration produced in a body is directly proportional to the applied force and inversely proportional to its mass.
- **Given**:
 - ✓ Force (F) = 250 N
 - ✓ Mass (m) = 50 kg
 - ✓ Initial velocity (u) = 0 m/s (the body starts from rest)
 - ✓ Time (t) = 8 s
- Using **Newton's Second Law**:
 - ✓ $F = ma$, therefore,
 - ✓ $a = F/m = 250/50 = 5 \text{ m/s}^2$
- Since the body starts from rest and moves with **uniform acceleration**, the **distance travelled is calculated using the second equation of motion**:
 - ✓ $s = ut + \frac{1}{2}at^2$
 - ✓ **Substituting the given values**:
 - ✓ $s = (0 \times 8) + \frac{1}{2} \times 5 \times (8)^2$
 - ✓ $s = 2.5 \times 64$
 - ✓ $s = 160 \text{ m}$
- Hence, the body covers a distance of **160 m** in **8 seconds**.

Q 20. Ans. C

Explanation:

- The **electrical energy consumed by an appliance is given by**:
 - ✓ **Electrical Energy = Power $\times$ Time**
 - ✓ First, calculate the power of the heater.
 - ✓ Power,

- ✓ $P = VI$
- ✓ $P = 220 \times 5 = 1100 \text{ W} = 1.1 \text{ kW}$
- ✓ Now convert the time into hours.
- ✓ $20 \text{ minutes} = 20/60 = 1/3 \text{ hour} = 0.333 \text{ hour}$

- Electrical energy consumed,
 - ✓ $\text{Energy} = 1.1 \times 0.333$
 - ✓ $\text{Energy} = 0.3667 \text{ kWh} \approx 0.367 \text{ kWh}$
- Hence, the electrical energy consumed by the heater in 20 minutes is 0.367 kWh.

Q 21. Ans. B

Explanation:

- When resistors are connected in **series**, the equivalent resistance is equal to the sum of their individual resistances.
- **Given:**
 - ✓ Resistance, $R_1 = 3 \Omega$
 - ✓ Resistance, $R_2 = 6 \Omega$
 - ✓ Resistance, $R_3 = 9 \Omega$
 - ✓ Voltage, $V = 36 \text{ V}$
- Total resistance,
 - ✓ $R = R_1 + R_2 + R_3$
 - ✓ $R = 3 + 6 + 9 = 18 \Omega$
- Now, applying **Ohm's Law**,
 - ✓ $I = V/R$
 - ✓ Substituting the values,
 - ✓ $I = 36/18$
 - ✓ $I = 2 \text{ A}$

Hence, the current flowing through the circuit is 2 A.

Q 22. Ans. A

Explanation:

- A convex mirror always forms a **virtual, erect, and diminished image**, irrespective of the object's position. Hence, it is used as a rear-view mirror in vehicles because it provides a wider field of view.
- A plane mirror always produces a **virtual, erect image** that is **the same size as the object** and is formed at the same distance behind the mirror as the object is in front of it.
- When an object is placed **beyond twice the focal length (2F)** of a convex lens, the image is formed **between F and 2F** on the opposite side. The image is **real, inverted, and diminished**.
- When an object is placed at the **centre of curvature (C)** of a concave mirror, the image is formed at C, and it is **real, inverted, and of the same size**. Since the image is always real, the best matching description is "Always real image."

Q 23. Ans. D

Explanation:

- **Pair 1 is correct:** Walking requires sufficient **static friction** between the feet and the ground. Ice provides very little friction, reducing grip and making walking difficult. Hence, this pair is correctly matched.
- **Pair 2 is correct:** Writing with a pencil or pen is possible because **friction between the writing tip and the paper** allows graphite or ink to be deposited on the paper. The contact involves **kinetic or sliding friction**, preventing slipping during writing.
- **Pair 3 is correct:** When a matchstick is rubbed against the striking surface, **friction produces heat**, raising the temperature of the chemicals on the match head to their ignition point, causing the matchstick to catch fire.
- **Pair 4 is correct:** Tyres are provided with **treads (grooves)** to increase friction between the tyre and the road surface. This improves grip, prevents skidding, and enhances vehicle safety, especially on wet roads.

Q 24. Ans. D

Explanation:

- **Pair 1 is correct:** A rainbow is formed when sunlight enters raindrops and undergoes **refraction, dispersion into constituent colours, internal reflection**, and finally refraction again while emerging from the raindrop. Thus, multiple internal reflection along with dispersion is responsible for rainbow formation.
- **Pair 2 is correct:** Stars appear to twinkle because their light passes through atmospheric layers of varying density and refractive index. This causes continuous **atmospheric refraction**, making the apparent position and brightness of stars fluctuate.
- **Pair 3 is correct:** A mirage occurs due to **refraction (and ultimately total internal reflection)** of light through layers of air having different temperatures and hence different refractive indices. The continuous refraction through these layers creates the illusion of water on a hot road or desert.
- **Pair 4 is correct:** A diamond has a very **high refractive index**, giving it a very small critical angle. Light entering the diamond undergoes repeated **total internal reflection**, producing its characteristic brilliance and sparkle.

Q 25. Ans. B

Explanation:

- The **First Law of Thermodynamics** introduces the concept of **internal energy**. It is based on the

principle of **conservation of energy**, stating that energy can neither be created nor destroyed but only transformed from one form to another.

- The law states that the heat supplied to a system is used partly to **increase its internal energy** and partly to **perform work**.

Q 26. Ans. A

Explanation:

- The **Second Law of Thermodynamics** states that **heat flows spontaneously from a body at a higher temperature to a body at a lower temperature**. The reverse process does not occur naturally and requires the expenditure of external work, as in refrigerators and air conditioners. This statement is known as the **Clausius statement** of the Second Law of Thermodynamics. Thus, heat naturally flows from a **hot body to a cold body** until thermal equilibrium is reached.

Q 27. Ans. B

Explanation:

- **Agnikul Cosmos**, a Chennai-based space startup, successfully launched the **Agnibaan SORTeD (SubOrbital Technological Demonstrator)** rocket on 30 May 2024. The mission marked the **world's first flight powered by a completely single-piece 3D-printed semi-cryogenic engine**, known as **Agnilet**. The launch was conducted from 'Dhanush', India's **first privately developed launchpad**, established by Agnikul Cosmos within the ISRO campus at Sriharikota, Andhra Pradesh. The **Agnilet engine**, manufactured as a **single 3D-printed component**, significantly reduces welding, assembly time, and production costs. As a **technology demonstration mission**, SORTeD was designed to validate the startup's **indigenous flight systems, avionics, flight computer, and other homegrown technologies** through in-flight testing. **Agnibaan** is a **customizable two-stage launch vehicle** capable of carrying payloads of up to 300 kg into a 700 km orbit.

Q 28. Ans. A

Explanation:

- The **PSLV-C62 mission** was launched by ISRO on 12 January 2026 from the **Satish Dhawan Space Centre, Sriharikota**. During the flight, the launch vehicle experienced an **anomaly in its third stage**, causing it to deviate from its planned trajectory and fail to place its payloads into the intended

orbit. The mission's primary payload was **EOS-N1 (Anvesha)**, an **Earth observation satellite**. In addition, the rocket carried **15 secondary satellites and technology demonstrators** from Indian and international startups and academic institutions, including **AayulSAT**, India's **first on-orbit satellite refuelling payload**.

Q 29. Ans. C

Explanation:

- **Kunchala Kyvalya Reddy** is a young Indian space enthusiast and **astronaut candidate** from **Nidadavole** in the **East Godavari district of Andhra Pradesh**. She has earned international recognition for her outstanding contributions to **astronomy** and **citizen science** at a remarkably young age. In 2025, at the age of 17, she was selected as an **Astronaut Candidate (ASCAN)** for the **Space Class 2025–2029 Astronaut Programme** conducted by **Titans Space Industries Inc. (TSI)**, a **Florida-based aerospace company**. As part of the four-year training programme, she will undergo **zero-gravity flights**, tactical training, and other preparations for future orbital missions. Earlier, as a **NASA citizen scientist** associated with the **International Astronomical Search Collaboration (IASC)**, she discovered two provisional **main-belt asteroids**—**2020 PS24** at the age of 12 and **2021 CM37** at the age of 14.

Q 30. Ans. D

Explanation:

- **Statement 1 is incorrect:** The **Surya AI model** is an **open-source AI foundation model** for solar physics and heliophysics, but it was developed by **NASA in collaboration with IBM, not ISRO**. The model was built to analyse the Sun's behaviour and improve the prediction of hazardous space weather events such as **solar flares** and **coronal mass ejections (CMEs)**.
- **Statement 2 is incorrect:** **Surya** is **not limited** to forecasting solar flares. In addition to predicting the **location, intensity, and shape of solar flares up to two hours in advance**, it can also perform several other heliophysics tasks, including:
 - ✓ **Active region segmentation**
 - ✓ **Solar wind speed estimation**
 - ✓ **Extreme Ultraviolet (EUV) spectral modelling**
 - ✓ **Analysis of broader solar dynamics and space weather.**
- **Statement 3 is incorrect:** **Surya** was **not trained on ground-based observations**. Instead, it was trained

using nine years of full-resolution data collected by NASA's Solar Dynamics Observatory (SDO), a space-based solar observatory that continuously monitors the Sun across multiple wavelengths.

Q 31. Ans. D

Explanation:

- Bihar's first dedicated Blackbuck Conservation Reserve is being developed in the Buxar district to protect the endangered blackbuck (kala hiran). The project is located in Baralew (Pilapur) Mauja of Navanagar block, where blackbucks inhabit the surrounding 400-acre forest area. As part of the initiative, a 12-acre rescue and conservation centre is being established to strengthen habitat protection, facilitate wildlife management, and promote eco-tourism in the region.

Q 32. Ans. C

Explanation:

- Nitish Kumar officially resigned as the Chief Minister of Bihar on 14 April 2026 after being elected to the Rajya Sabha, marking his transition to national politics. He submitted his resignation to the Bihar Governor, Lt. Gen. (Retd.) Syed Ata Hasnain at Lok Bhavan, Patna. His resignation brought to a close two decades of leadership in the state, following which Samrat Choudhary was sworn in as the new Chief Minister of Bihar.

Q 33. Ans. B

Explanation:

- The official mascot of the Men's Hero Asia Cup 2025, which was held in Rajgir, Bihar, was Chaand, a tiger inspired by India's national animal and the majestic tigers of Valmiki Tiger Reserve. The mascot's name pays tribute to hockey legend Major Dhyan Chand, reflecting his legendary dedication to practising under the moonlight. Chaand's distinctive appearance includes a magician's hat, symbolising Dhyan Chand's extraordinary stickwork, and a red cape, representing courage, strength, and sporting excellence.

Q 34. Ans. C

Explanation:

- Gross State Domestic Product (GSDP) refers to the total monetary value of all final goods and services produced within a state during a particular financial year.
- According to the Bihar Budget 2026–27, Bihar's

GSDP at current prices for 2026–27 is estimated at ₹13.1 lakh crore.

- The projection reflects the expected expansion of Bihar's economy due to growth in:
 - ✓ Agriculture and allied activities
 - ✓ Infrastructure development
 - ✓ Manufacturing activities
 - ✓ Service-sector expansion

Q 35. Ans. C

Explanation:

- According to the Bihar Economic Survey 2025–26, Bihar's per capita GSDP at current prices for 2024–25 is estimated at ₹76,490. Per capita GSDP is calculated by dividing the total Gross State Domestic Product (GSDP) of a state by its total population and indicates the average economic output per person. It is an important indicator used to assess the economic development level of a state and compare income levels across states. However, it does not represent the actual income earned by every individual. For comparison, India's per capita GDP in 2024–25 was around ₹2,34,859, which is much higher than Bihar's per capita GSDP.

Q 36. Ans. B

Explanation:

- Revenue Balance = Revenue Receipts – Revenue Expenditure
- It shows whether the government's regular income is sufficient to meet its regular expenses.
 - ✓ Revenue Surplus: When revenue receipts > revenue expenditure, the government has extra money after meeting day-to-day expenses.
 - ✓ Revenue Deficit: When revenue expenditure > revenue receipts, the government needs to borrow even for routine expenses.
- According to the Bihar Budget 2026–27 estimates:
 - ✓ Bihar is expected to achieve a revenue surplus of ₹1,143 crore. This is approximately 0.1% of Bihar's GSDP.

Q 37. Ans. B

Explanation:

- According to the India State of Forest Report (ISFR) 2023, Bihar's total forest and tree cover is 9,902.66 sq. km, which accounts for approximately 10.52% of the state's geographical area. The green cover of Bihar consists of both forest cover and tree cover. The forest cover constitutes 7,532.45 sq. km (8.00% of the geographical area), while the tree

cover accounts for 2,370.21 sq. km (around 2.52% of the geographical area).

- Under the classification of forest density, Bihar's forest cover is divided into three categories. **Very Dense Forest (VDF)** covers 333.42 sq. km, **Moderately Dense Forest (MDF)** covers 3,285.83 sq. km, and **Open Forest (OF)** constitutes the largest share with 3,913.20 sq. km.

Q 38. Ans. D

Explanation:

- **Statement 1 is correct:** Dr. Rajendra Prasad played an important role in the **Champaran Satyagraha of 1917**, which was the first major Satyagraha movement launched by **Mahatma Gandhi in India**. The movement was organised against the exploitation of indigo farmers by British planters under the **Tinkathia system**. Rajendra Prasad, along with other leaders like **Acharya J. B. Kripalani and Brajkishore Prasad**, supported Gandhi's efforts in Champaran. This experience deeply influenced him and brought him closer to Gandhian ideas.
- **Statement 2 is correct:** Dr. Rajendra Prasad wrote his autobiography titled "**Atmakatha**", which describes his early life, education, legal career, association with the Indian freedom movement, and experiences with leaders like Mahatma Gandhi. It provides important insights into the nationalist movement from his perspective.
- **Statement 3 is correct:** Dr. Rajendra Prasad became the **first President of India on 26 January 1950** and served as President until **13 May 1962**. He was elected for a second term in 1957, making him the **only President of India to complete two full terms** in office. No other Indian President has served two complete terms.

Q 39. Ans. A

Explanation:

- **Statement 1 is incorrect:** Dr. Ram Manohar Lohia was born on **23 March 1910 in Akbarpur, Uttar Pradesh**, not in the Siwan district of Bihar. He belonged to Uttar Pradesh and later became one of India's most influential socialist thinkers and political leaders.
- **Statement 2 is correct:** Dr. Ram Manohar Lohia played a major role in the formation of the **Socialist Party in December 1955**. He established the party after resigning from the post of **General Secretary of the Praja Socialist Party (PSP)** due to ideological differences.

- ✓ His political ideology was based on democratic socialism, which focused on:

- Social equality
- Decentralisation of power
- Removal of economic and social inequalities
- Opposition to both Western capitalism and Soviet-style communism

Q 40. Ans. B

Explanation:

- The Ganga River flows from west to east through Bihar, dividing the state into North Bihar and South Bihar. The tributaries joining the Ganga from the northern side are called **left-bank tributaries**, while those joining from the southern side are called **right-bank tributaries**.
- **Left-Bank (Northern) Tributaries of the Ganga:**
 - ✓ The **Gandak River** is an important **left-bank tributary of the Ganga**. It originates in the **Nepal Himalayas** and enters Bihar before joining the Ganga near **Patna (near Hajipur/Sonpur region)**. It is one of the major rivers of North Bihar.
 - ✓ The **Bagmati River** is a **left-bank tributary system of the Ganga**. It originates in the Himalayan region of Nepal and flows through the plains of North Bihar. It is associated with frequent floods in Bihar and ultimately becomes part of the larger Kosi river system before reaching the Ganga.
 - ✓ The **Mahananda River** is a **left-bank tributary of the Ganga**. It originates in the **Darjeeling hills** and flows through northeastern Bihar before joining the Ganga system in West Bengal.
- ✓ **Right-Bank (Southern) Tributaries of the Ganga:**
 - The **Kiul-Harohar river system** is a **right-bank tributary system of the Ganga**. It originates in the Jharkhand region and drains parts of southern Bihar before joining the Ganga. Therefore, it is not a left-bank tributary.
 - The **Chandan River** is also a **right-bank tributary of the Ganga**. It originates from the hilly regions of Jharkhand and flows through southeastern Bihar before joining the Ganga system.

Q 41. Ans. D

Explanation:

- **Assertion (A) is False:** Agro-Climatic Zone III of Bihar (comprising Zone IIIA and IIIB) is considered

one of the regions vulnerable to **drought conditions** due to comparatively lower and uneven rainfall distribution, frequent dry spells, and dependence on monsoon rainfall in several districts. The **Government of Bihar's drought assessment and agricultural planning documents** identify several districts of South Bihar as drought-prone areas. Therefore, the statement that Zone III is generally not treated as drought susceptible is incorrect.

- **Reason (R) is true:** Agro-Climatic Zone III represents the **South Bihar region** and is divided into:
 - ✓ **Zone IIIA – South-East Alluvial Plain**
 - ✓ **Zone IIIB – South-West Alluvial Plain**
 - ✓ Together, these zones include multiple districts of southern Bihar, such as Bhagalpur, Banka, Munger, Jamui, Lakhisarai, Sheikhpura, Patna, Gaya, Rohtas, Bhojpur, Buxar, Aurangabad, and others.

Q 42. Ans. D

Explanation:

- **Assertion (A) is False:** cropping patterns and crop recommendations vary across different agro-climatic regions based on **rainfall, soil type, temperature, topography, and local environmental conditions**. Even within the same agro-climatic zone, variations in local climate, irrigation availability, flood-prone or drought-prone conditions, and soil characteristics influence the choice of crops and varieties. Therefore, local microclimatic conditions do have a significant impact on cropping decisions.
- **Reason (R) is true:** The Government of Bihar divides the state into four agro-climatic zones based on factors such as **soil characteristics, rainfall pattern, topography (terrain), and climatic conditions**. This zoning helps in recommending suitable crops, varieties, and agricultural practices for different regions, thereby improving agricultural productivity.

Q 43. Ans. D

Explanation:

- The **Sakri–Nata River Interlinking Project** is a **proposed water-transfer project in South Bihar**, designed to transfer surplus monsoon water from the **Sakri River** to the **Nata River**. The primary objective is to improve irrigation, reduce water scarcity in drought-prone areas, and ensure better utilisation of excess monsoon runoff.
- **Statement 1 is incorrect:** The project is **not located**

in North Bihar. It is proposed for **South Bihar**, a region that frequently experiences **dry spells and drought-like conditions**, particularly during the summer months. The project aims to divert surplus water from the Sakri River, which receives abundant monsoon flow, to the comparatively water-deficient Nata River basin.

- **Statement 2 is incorrect:** The project is **not intended for these North Bihar districts**. Instead, it is expected to provide irrigation benefits to farmers in the **Nawada, Nalanda, Sheikhpura, and Jamui districts of South Bihar**. Once completed, it is expected to **irrigate more than 68,800 hectares of agricultural land**, improving agricultural productivity and water availability in these drought-prone areas.

Q 44. Ans. B

Explanation:

- The **official State Flower of Bihar is the Marigold (Genda Phool)**. The Bihar Government officially declared Marigold as the State Flower in April 2013 to promote its preservation and cultural significance. The flower was selected because of its **wide cultivation in Bihar, vibrant orange and yellow colour, association with prosperity, and extensive use in religious ceremonies, festivals, and social occasions**. Marigolds belong to the **genus Tagetes**, with common varieties including **African Marigold (Tagetes erecta)** and other cultivated forms.

Q 45. Ans. D

Explanation:

- The **Individual Satyagraha** was initiated by **Mahatma Gandhi in October 1940** to protest against the British decision to involve India in **World War II without obtaining the consent of Indian leaders**. The movement aimed to uphold the right of Indians to express their opposition to British policies through non-violent methods. In **Bihar**, the movement began soon after its national launch and witnessed participation from prominent political leaders as well as women activists. The Bihar phase was formally led by **Sri Krishna Singh**, who became the **first Satyagrahi from Bihar**. **Dr. Anugrah Narayan Sinha** was recognised as the **second prominent Satyagrahi from Bihar**. Another important participant, **Gauri Shankar Singh**, performed Satyagraha at **Gaya on 4 December 1940**. The movement also witnessed significant participation of women activists such as **Priyambada Devi, Janki Devi, Jagatrani Devi, and**

Mrs. Mahadevi Kejariwal, who actively spread anti-British and anti-war messages. Many participants, including women activists, faced arrest, especially in regions like Gaya, highlighting the widespread nature of the Individual Satyagraha movement in Bihar.

Q 46. Ans. B

Explanation:

- **Sujni (Sujani) Embroidery**, a traditional embroidery craft of Bihar, received the **Geographical Indication (GI) tag in 2006**. Practised primarily in the Bhusra village of Muzaffarpur district and selected villages of Madhubani district, this unique handicraft is known for its intricate needlework and storytelling motifs. The GI registration under the Geographical Indications of Goods (Registration and Protection) Act, 1999, provides legal protection to the craft and its artisans.

Q 47. Ans. A

Explanation:

- **Statement 1 is correct:** The Kosi River is known as **Saptakoshi in Nepal**, as it is formed by the confluence of **seven major tributaries/rivers originating from the Himalayan region**. These rivers include the Arun, Sun Kosi, Tamur and their tributaries. After the confluence in Nepal, the river enters India through Bihar and flows into the Ganga. Therefore, Statement 1 is correct.
- **Statement 2 is correct:** The Kosi River basin covers a wide range of altitudes, extending from the **high Himalayan region in the north**, where climatic conditions are **frigid**, to the **Gangetic plains in the south**, where the climate is **tropical/subtropical**. This variation in altitude results in diverse climatic and ecological conditions within the basin. Therefore, Statement 2 is correct.
- **Statement 3 is incorrect:** The Kosi River is classified as an **antecedent river**, not a consequent river. Its headwaters are considered older than the Himalayan uplift. As the Himalayas rose, the river maintained its course by cutting through the rising mountain ranges and later formed a large alluvial fan in the Gangetic plains. Therefore, Statement 3 is incorrect.

Q 48. Ans. D

Explanation:

- The **current Chairman of the Bihar Legislative Council (Vidhan Parishad)** is **Awadhesh Narain Singh**, who became the **13th Chairman** after

assuming office on **20 June 2024**. A senior leader of the **Bharatiya Janata Party (BJP)**, he presides over the proceedings of the Council, while **Prof. Ram Bachan Rai (JD-U)** serves as the **Deputy Chairperson**. The **Bihar Legislative Council** is the **upper house** of Bihar's bicameral legislature. It is a **permanent House** comprising **75 members**, each serving a **six-year term**, with **one-third of the members retiring every two years**, ensuring continuity in the legislative process.

Q 49. Ans. B

Explanation:

- The **2026 Mandar Mahotsav (Bounsia Mela)** was celebrated in **Banka district, Bihar**, from **14 to 18 January 2026**, beginning on the auspicious occasion of **Makar Sankranti**. The five-day festival showcased Bihar's rich religious, cultural, and artistic heritage. Major attractions included the traditional **Ratha Yatra of Lord Madhusudana** and celebrations linked to the legend of the **Samudra Manthan (Churning of the Ocean)** associated with the sacred **Mandar Hill**. The event also featured performances by renowned singers such as **Wasir Desai, Kavita Paudwal, Sumi Sree, and Sawan Savre**, along with **agricultural exhibitions, local handicraft displays, and Kavi Sammelan (poetry gatherings)**, making it one of Bihar's prominent cultural festivals.

Q 50. Ans. B

Explanation:

- **Shailesh Kumar** of India scripted history by winning the country's **first-ever gold medal** at the **World Para Athletics Championships 2025**. Hailing from **Islam Nagar village in Jamui district, Bihar**, he secured the gold medal in the **men's high jump T63 event** by clearing a **personal best of 1.91 metres**, setting both a **Championship Record** and an **Asian Record**. Fellow Indian **Varun Bhati** won the **bronze medal** in the same event. The achievement marked **Shailesh Kumar's maiden World Championship gold** and highlighted India's rising prominence in international para athletics.

Q 51. Ans. C

Explanation:

- The **Tolkappiyam**, attributed to the grammarian **Tolkappiyar**, is the **oldest extant Tamil grammatical treatise** and the **foundational text** for the study of ancient Tamil grammar (phonology, morphology, syntax) as well as poetics (covering

the classification of poetic themes, genres, and literary conventions used in Sangam literature). It is traditionally divided into three parts: Ezhuthu (phonology/orthography), Sol (morphology/grammar of words), and Porul (subject matter, dealing with poetic conventions, love, war, ethics, and polity).

● **Additional facts:**

- ✓ **Silappatikaram:** One of the “Five Great Epics” of Tamil literature, composed by Ilango Adigal, telling the story of Kannagi and Kovalan, a work of narrative literature, not a grammatical/poetic treatise.
- ✓ **Manimekalai:** Another of the Five Great Epics, composed by Sattanar, considered a sequel to the Silappatikaram, dealing with Buddhist philosophy, again literary, not grammatical.
- ✓ **Pathruppattu:** One of the “Ten Idylls” (Pattupattu) of Sangam literature, comprising ten poems praising various Chera kings, a work of Sangam poetry itself, not a text on grammar/poetics.

Q 52. Ans. A

Explanation:

- **Statement 1 is correct:** The Naneghat inscription, issued by Queen Naganika (wife of Satakarni I), is generally regarded as the earliest epigraphic evidence of land grants, it records villages being offered as dakshina (sacrificial gifts) to Brahmin priests during Vedic sacrifices performed by her husband. This makes the Satavahanas the first Indian dynasty on record to institute the practice of royal land grants.
- **Statement 2 is correct:** Later Satavahana land grants, most notably recorded in the Nasik cave inscriptions of Gautamiputra Satakarni, explicitly came with fiscal and administrative immunities (parihara), such grants specified that the land was not to be entered by royal troops, not to be dug for salt, and was free from the control of state officials, effectively exempting it from both taxation and administrative interference. This is considered a landmark development by historians (like R.S. Sharma) in tracing the roots of Indian feudalism.
- **Statement 3 is incorrect:** Despite being a wealthy trading dynasty, the Satavahanas did not issue significant gold coinage. Their coins were predominantly made of lead, potin (copper-lead-tin-zinc alloy), copper, and bronze, with silver coins appearing more regularly only after Gautamiputra

Satakarni (largely through restriking captured Western Kshatrapa silver coins, such as those of Nahapana). Gold was known and valued as a metal, but the Satavahanas are notably not associated with any “massive volume of high-purity imperial gold coins”, that description instead fits dynasties like the Guptas or Kushans

Q 53. Ans. C

Explanation:

- **Pair 1 is correct:** The Karle Chaitya (the largest rock-cut chaitya in India) was heavily patronised by wealthy merchant guilds (including Yavana or Greek traders) and received significant land and revenue grants from Satavahana rulers (like Gautamiputra Satakarni) and the Kshatrapa ruler Nahapana.
- **Pair 2 is correct:** The Naneghat cave inscriptions were commissioned by Queen Nayanika (Naganika), wife of the early Satavahana king Satakarni I. The inscription is a major historical record detailing their performance of grand Vedic sacrifices (like Ashvamedha and Rajasuya) and massive state donations.
- **Pair 3 is correct:** Kanheri Caves boast over 100 inscriptions detailing continuous patronage from a wide demographic, including wealthy merchants, royal female donors, and, uniquely, Buddhist monks and nuns themselves who acted as prominent donors.
- **Additional Facts:**
 - ✓ **Karle's Woodwork:** The Karle Chaitya is architecturally famous for retaining its original 2,000-year-old curved teakwood ribs on its vaulted ceiling, as well as its highly polished lion pillars.
 - ✓ **Naneghat Trade Route:** Naneghat was not just a religious site; it was a major ancient tollgate and trade pass connecting the Konkan coast ports (like Kalyan and Sopara) to the capital of the Deccan plateau (Junnar).
 - ✓ **Kanheri's Longevity:** Unlike most rock-cut sites that were abandoned early, Kanheri remained a continuously active Buddhist monastic settlement from the 1st century BCE all the way up to the 11th century CE.

Q 54. Ans. C

Explanation:

- **Statement 1 is correct:** Kushana kings adopted grand, divine-sounding titles to legitimise their

authority. The most famous is Devaputra (Son of God), a concept they directly borrowed from the Chinese imperial title Tianzi (Son of Heaven).

- **Statement 2 is correct:** Emperor Kanishka's accession to the throne in 78 CE marks the beginning of the Shaka era (Saka Samvat). Despite being founded by a Kushana king, it is called the Shaka era because it was widely used by the Shaka satraps of Ujjain for centuries. It is now the official national calendar of India.

- **Additional Facts:**

- ✓ **The Satrap System:** To manage their vast, multi-ethnic empire, the Kushanas utilised the Kshatrapa (Satrap) system, where the empire was divided into provinces ruled by military governors.
- ✓ **Dual Capitals:** The empire had two major administrative hubs: **Purushapura** (modern Peshawar) serving as the primary capital, and **Mathura** functioning as a major secondary capital and cultural centre.
- ✓ **Monetary Revolution:** The Kushanas were the first dynasty in the Indian subcontinent to issue gold coins on a massive scale (the Dinar), a testament to the immense wealth generated by their control over the Silk Route.
- ✓ **Cultural Syncretism:** Kushana coinage is famous for its extreme religious tolerance, depicting a remarkable blend of Greek, Roman, Iranian, Hindu, and Buddhist deities (including Shiva, Buddha, and Heracles) on the reverse sides of their coins.

Q 55. Ans. B

Explanation:

- The Kushanas (along with the Shakas) popularised the **Satrapy (Kshatrapa) system** in India for provincial administration. Because their empire was vast and multi-ethnic, it was divided into provinces ruled by military governors known as Mahakshatrapas (Great Satraps) or Kshatrapas (Satraps).
- **Iqta** was introduced much later by the Delhi Sultanate (Iltutmish).
- **Mandala** (as a provincial unit) was characteristic of the Chola Empire (Mandalam).
- **Ashtapradhan** was the council of eight ministers established by the Maratha king Shivaji.
- **Additional Facts:**
 - ✓ **Dual Kingship (Dvairajya):** A unique feature of Kushana governance was the practice of

joint rule, where a senior king and a junior king (often father and son) shared power simultaneously to ensure smooth succession and firm control over vast territories.

- ✓ **Military-Centric Bureaucracy:** They introduced high-sounding military-administrative titles like Dandanayaka (commander/magistrate) and Mahadandanayaka, which later became foundational elements of the Gupta administrative framework.
- ✓ **Feudatory Autonomy:** Because the empire stretched from Central Asia to Pataliputra, the Kshatrapas enjoyed a high degree of regional autonomy, operating almost as vassal kings, as long as they paid tribute to the central "King of Kings" (Maharajadhiraja).

Q 56. Ans. C

Explanation:

- **Nyaya (a - 1):** Founded by sage Gautama, this school is centred on logic, epistemology, and the rules of debate. It states that obtaining valid knowledge (Pramana) is the only way to attain liberation.
- **Vaisheshika (b - 2):** Founded by sage Kanada, this school is known for its physical sciences and atomic theory. It postulates that all objects in the physical universe are reducible to a finite number of indivisible atoms (Paramanu).
- **Samkhya (c - 3):** Founded by sage Kapila, it is considered the oldest philosophical school. It is strongly dualistic, dividing reality into Purusha (consciousness) and Prakriti (matter).
- **Mimamsa (d - 4):** Founded by sage Jaimini (also known as Purva Mimamsa), this school focuses heavily on the philosophical justification of Vedic rituals, duty (Dharma), and the ultimate authority of the Vedas.
- **Additional Facts:**
 - ✓ **The Missing Two:** The remaining two schools of the six Astika (orthodox) philosophies are **Yoga** (founded by Patanjali, focusing on meditation and physical/mental discipline) and **Vedanta / Uttara Mimamsa** (founded by Badarayana, focusing on the metaphysical teachings of the Upanishads).
 - ✓ **Astika vs. Nastika:** These six schools are Astika because they accept the supreme authority of the Vedas. In contrast, Nastika (heterodox) schools—such as Buddhism, Jainism, and Charvaka (Lokayata)—completely reject the authority of the Vedas.

Q 57. Ans. C

Explanation:

- **Statement 1 is correct:** The Harappan weighing system was highly standardised. For lower denominations, the weights followed a strict binary system (1, 2, 4, 8, 16, 32, etc., up to 12,800). For higher denominations, they utilised a decimal system.
- **Statement 2 is correct:** These standard weights were primarily made of a hard stone called **chert**. They were characteristically cubical in shape and devoid of any markings.

Additional Facts:

- ✓ **Uniform Brick Ratio:** Standardisation was not limited to weights. Across the entire Indus Valley Civilisation, burnt bricks were completely standardised with a strict length-to-breadth-to-thickness ratio of 4:2:1.
- ✓ **Linear Measurements:** Harappans also developed precise tools for measuring length. Archaeological finds include a highly accurate ivory scale discovered at Lothal and a shell scale found at Mohenjo-Daro.
- ✓ **Seals:** Harappan seals, used for trade and administrative purposes, were also highly standardised. They were mostly square-shaped and uniformly crafted from **steatite** (a soft soapstone).

Q 58. Ans. C

Explanation:

- The Harappan script was often written in a **Boustrophedon** style. This means the writing goes alternately from right-to-left on the first line, then left-to-right on the second line, and so on (like an ox ploughing a field).
- **Cuneiform** is the wedge-shaped script used by the ancient Mesopotamians.
- **Hieroglyphics** are the pictographic writing system of ancient Egypt.
- **Brahmi** is an ancient Indian script (deciphered by James Prinsep) used for Ashokan edicts, written strictly left-to-right.
- **Additional Facts:**
 - ✓ **General Direction:** While boustrophedon was used for multi-line texts, the general direction of single-line Harappan inscriptions was **right-to-left**. This is evident because the symbols are often crowded on the left side of the seals, indicating the writer started on the right and ran out of space.

- ✓ **Undeciphered:** To this day, the Harappan script remains undeciphered, which is why we cannot read their literature or administrative records.
- ✓ **Logographic Nature:** The script is not alphabetical. It is predominantly pictographic or logo-syllabic, meaning each symbol represents a word, idea, or syllable. There are roughly 400 to 600 distinct signs found across Harappan sites.
- ✓ **Longest Inscription:** The longest single Harappan inscription found to date contains about 26 signs. Additionally, the famous “Dholavira Signboard” consists of 10 large signs originally inlaid in white gypsum.

Q 59. Ans. B

Explanation:

- **Statement 1 is incorrect:** The Harappan weight system did not fail to influence later periods; rather, it showed remarkable continuity. Metrologists and historians (like D.D. Kosambi) have demonstrated that the weight standards of India’s earliest historical punch-marked coins (from the Mahajanapada and Mauryan periods) strictly correspond to the traditional Harappan weight metrics based on the ratti (gunja) seed.
- **Statement 2 is correct:** Many religious motifs from the Indus Valley Civilisation persisted and evolved into core elements of later Hinduism. These include the veneration of the pipal tree, the use of the Swastika symbol, water purification rituals (evidenced by the Great Bath), and the worship of a male deity depicted on seals in a yogic posture surrounded by animals (Proto-Shiva / Pashupati).
- **Statement 3 is correct:** The Harappan standardized brick ratio of 4:2:1 (length:breadth:thickness) is structurally optimal for bonding. Because of its structural integrity, this specific mathematical ratio continued to be the standard for baked-brick construction in the Indian subcontinent long after the decline of the IVC.
- **Additional Facts:**
 - ✓ **Lost-Wax Casting:** The cire-perdue (lost wax) technique used by Harappans to cast the famous bronze “Dancing Girl” of Mohenjo-Daro remained the standard method for Indian metallurgy for millennia, reaching its peak in the Chola bronze sculptures.
 - ✓ **Cotton Cultivation:** The Greeks called cotton Sindon because it originated in the Sindh

(Indus) region. The Harappans were among the first in the world to spin and weave cotton, laying the foundation for India's historic textile industry.

- ✓ **Urban Planning:** While their sophisticated underground drainage systems and grid-planned cities disappeared for a few centuries during the Vedic period, these urban concepts re-emerged during the Second Urbanisation in the Gangetic plains (around the 6th century BCE).

Q 60. Ans. A

Explanation:

- **Statement 1 is correct:** The commercialisation of agriculture under British rule forced a massive shift from traditional subsistence farming (food crops like wheat and rice) to the cultivation of cash crops (like indigo, cotton, jute, tea, sugarcane, and opium) intended primarily for export to feed British industries.
- **Statement 2 is incorrect:** The process completely devastated the average Indian peasant. It was a forced commercialisation driven by the need to pay high land revenue in cash. Peasants fell into severe debt traps laid by moneylenders, lost their land, and suffered through catastrophic famines because food crop production plummeted.
- **Additional Facts:**
 - ✓ **The Cotton Boom & Bust:** During the American Civil War (1861–1865), British textile mills lost access to American cotton, leading to a massive boom in Indian cotton exports. However, when the war ended, and American cotton returned, Indian prices crashed, leaving peasants in the Deccan region in insurmountable debt (leading to the Deccan Riots of 1875).
 - ✓ **The Tinkathia System:** A notorious example of forced commercialisation was the Tinkathia system in Champaran, Bihar, where European planters legally bound peasants to grow indigo on 3/20th of their land holdings at highly exploitative rates.
 - ✓ **Impact on Food Security:** By shifting the best, most fertile lands to cash crop production and tying the rural economy to volatile global market prices, India became highly vulnerable to starvation. The late 19th century saw some of the most devastating famines in Indian history.

Q 61. Ans. A

Explanation:

- **Campbell Commission (a - 4):** Appointed in 1866 by Viceroy Lord Lawrence following the devastating Orissa Famine. It was the first serious attempt to investigate the causes of famine and recommend relief measures.
- **Strachey Commission (b - 2):** Appointed in 1878 by Viceroy Lord Lytton after the Great Famine of 1876–1878 (which severely affected Madras, Bombay, and Mysore). It recommended that the state should take the responsibility of providing relief and suggested creating a Famine Fund.
- **Lyall Commission (c - 3):** Appointed in 1897 by Viceroy Lord Elgin II to review the workings of the existing Famine Codes following the widespread famine of 1896–1897.
- **MacDonnell Commission (d - 1):** Appointed in 1900 by Viceroy Lord Curzon after the severe famine of 1899–1900. It stressed the “moral strategy” of advancing loans to peasants, early suspension of land revenue, and better irrigation networks.
- **Additional Facts:**
 - ✓ **The Famine Code (1883):** The most significant outcome of the Strachey Commission was the drafting of the Famine Code in 1883. It established standard protocols for declaring a famine, setting up relief works, providing gratuitous relief to the vulnerable, and suspending revenue collection.
 - ✓ **Woodhead Commission (1944):** The last major famine commission of British India, appointed by Lord Wavell to investigate the catastrophic Bengal Famine of 1943 (which was exacerbated by wartime policies and grain hoarding).
 - ✓ **The Famine Relief Fund:** Based on Strachey's recommendations, the British government introduced a Famine Insurance Grant, requiring a dedicated annual budget allocation to build railways and irrigation canals specifically for famine prevention.

Q 62. Ans. C

Explanation:

- The **Guarantee System** was introduced in the 1850s (notably under Lord Dalhousie) to attract private British investment for railway construction in India. Under this system, the British Indian government provided free land and guaranteed a minimum

return (usually 5%) on the capital invested by private British railway companies. Any shortfall in profits was made up from Indian revenues.

● **Additional Facts:**

- ✓ The primary goal of building railways was not the economic development of India, but rather colonial exploitation, facilitating the rapid export of raw materials (like cotton), the inland distribution of British manufactured goods, and the swift movement of troops to suppress rebellions.
- ✓ The Guarantee System became a massive burden on the Indian exchequer. Because the 5% return was guaranteed regardless of efficiency, private companies had no incentive to minimise construction costs, leading to highly inflated expenses paid for by Indian taxpayers.
- ✓ The first passenger railway line in India ran from Bombay to Thane in 1853, covering a distance of 34 kilometres.
- ✓ **Acworth Committee (1921):** This committee investigated railway management and recommended the nationalisation of Indian Railways. It also led to the separation of the Railway Budget from the General Budget in 1924 (a practice that continued until they were merged again in 2017).

Q 63. Ans. D

Explanation:

- **Pair 1 is correct:** The Santhal Rebellion (1855–56), also known as the Santhal Hul, took place in the Rajmahal Hills region (Damin-i-Koh, present-day Jharkhand). It was led by the brothers Sidhu and Kanhu against the oppressive Zamindari system and British revenue policies.
- **Pair 2 is correct:** The Munda Ulgulan (Great Tumult) of 1899–1900 took place in the Chhotanagpur plateau region (south of Ranchi). It was led by Birsa Munda to protest the erosion of their traditional Khuntkatti (joint landholding) system by jagirdars and thikadars.
- **Pair 3 is correct:** The Rampa Rebellion (1922–1924) occurred in the Godavari Agency tracts (present-day Andhra Pradesh). It was led by Alluri Sitarama Raju against the British Madras Forest Act of 1882, which restricted the tribal people's traditional podu (shifting) cultivation.
- **Pair 4 is correct:** The Ahom Revolt (1828) took place in the Assam region. Following the First Anglo-Burmese War (1824–26), the British pledged

to withdraw from Assam but instead attempted to incorporate it into their territory. This sparked a rebellion led by Gomdhar Konwar.

● **Additional Facts:**

- ✓ **Diku:** A common theme in the Chhotanagpur and Santhal rebellions was the expulsion of the Dikus, a term used by the tribal population for exploitative outsiders (moneylenders, traders, and British officials).
- ✓ **Chotanagpur Tenancy Act (1908):** A direct result of the Munda Ulgulan. The British were forced to pass this act to restrict the transfer of tribal land to non-tribals, legally recognising the Khuntkatti rights.
- ✓ **Santhal Parganas:** After the massive Santhal uprising, the British were forced to create a separate administrative district called the Santhal Parganas to pacify the region, applying special laws to protect tribal lands.

Q 64. Ans. D

Explanation:

- **Pair 1 is correct:** Nana Saheb (the adopted son of the last Peshwa, Baji Rao II) led the revolt in Kanpur. He was assisted by his brilliant commander, Tantia Tope, and Azimullah Khan.
- **Pair 2 is correct:** Begum Hazrat Mahal took over the leadership in Lucknow (Awadh). She declared her minor son, Birjis Qadr, as the Nawab and organised a fierce armed rebellion against the British following the controversial annexation of Awadh in 1856.
- **Pair 3 is correct:** Khan Bahadur Khan, a descendant of the former Rohilla rulers, led the uprising in Bareilly (Rohilkhand). He organised a strong army of 40,000 soldiers and offered stiff resistance to the British.
- **Pair 4 is correct:** Kunwar Singh, the nearly 80-year-old Zamindar of Jagdishpur, was the chief organiser of the revolt in Bihar. He successfully occupied Arrah and actively fought the British across multiple regions until his death.
- **Additional Facts:**
 - ✓ While the 82-year-old Mughal Emperor Bahadur Shah Zafar was declared the symbolic leader of the revolt, the actual military command in Delhi was held by a court of soldiers headed by **General Bakht Khan**.
 - ✓ **Faizabad:** The revolt in Faizabad was led by **Maulvi Ahmadullah**, who proved to be one of the most formidable enemies of the British, utilising guerrilla warfare tactics effectively.
 - ✓ Specific British commanders were responsible

for recapturing these centres. Sir Colin Campbell recaptured Kanpur and Lucknow, while Sir Hugh Rose suppressed the rebellion in Jhansi and Gwalior (famously calling Rani Lakshmibai the “only man among the rebels”).

- ✓ **The Royal Title Act (1876):** As a long-term consequence of the 1857 revolt and the subsequent transfer of power to the Crown (Government of India Act 1858), Queen Victoria formally assumed the title of Empress of India (Kaiser-i-Hind).

Q 65. Ans. A

Explanation:

- **Statement 1 is correct:** The Ghadar Party (originally named the Pacific Coast Hindustan Association) was founded in 1913 in San Francisco, USA, predominantly by Punjabi Sikh expatriates. Its headquarters were known as the Yugantar Ashram.
- **Statement 2 is correct:** The party published a weekly paper called Ghadar (meaning Revolt), carrying the caption “Angrezi Raj Ka Dushman” (Enemy of British Rule). It was published in Urdu, Gurmukhi, Hindi, and Gujarati to appeal to a broad Indian demographic.
- **Statement 3 is incorrect:** While they planned a massive, synchronised pan-India mutiny in the British Indian Army scheduled for February 21, 1915 (the Ghadar Mutiny), the plot failed. It was compromised by British intelligence through informants (like Kirpal Singh), leading to mass arrests and the execution of key leaders before the mutiny could begin.
- **Additional Facts:**
 - ✓ **Key Leaders:** Founded under the presidency of Sohan Singh Bhakna, with Lala Hardayal as the general secretary. Other prominent figures included Kartar Singh Sarabha, Barkatullah, and Vishnu Ganesh Pingle.
 - ✓ **Rash Behari Bose:** The famous revolutionary Rash Behari Bose was invited to lead the planned 1915 armed revolt in India. After the plot’s failure, he managed to escape to Japan, where he later helped establish the Indian Independence League.
 - ✓ **The Komagata Maru Incident (1914):** A major catalyst for the Ghadar movement. A Japanese steamship carrying 376 Indians was denied entry into Canada due to discriminatory exclusion laws. Upon being forced back to Calcutta, British police fired on the passengers

at Budge Budge, killing many and heavily radicalising the Indian diaspora.

- ✓ **Defence of India Act (1915):** The British passed this draconian wartime legislation specifically to crush the Ghadarites and other revolutionary activities, allowing for preventive detention and summary trials.

Q 66. Ans. B

Explanation:

- **Statement 1 is incorrect:** The primary political goal of the Nehru Report was **Dominion Status** for India within the British Commonwealth, not Complete Independence. This cautious demand actually caused a rift within the Congress, leading younger leaders like Jawaharlal Nehru and Subhas Chandra Bose to oppose the report and form the Independence for India League to push for Purna Swaraj (which was finally adopted in 1929 at the Lahore Session).
- **Statement 2 is correct:** The Nehru Report explicitly rejected the British system of separate communal electorates, arguing that it was divisive and hindered national unity. Instead, it recommended **joint electorates** with reserved seats for Muslims at the Centre and in provinces where they were in a minority.
- **Additional Facts:**
 - ✓ The Nehru Committee was formed as a direct response to a challenge thrown by Lord Birkenhead (the Secretary of State for India), who dared Indian politicians to draft a constitution that had the unanimous backing of all Indian political factions, following the boycott of the all-white Simon Commission.
 - ✓ The committee was chaired by **Motilal Nehru** (Jawaharlal Nehru’s father). Other prominent members included Tej Bahadur Sapru, Subhas Chandra Bose, and Ali Imam.
 - ✓ A highly progressive feature of the report was the inclusion of 19 Fundamental Rights, which guaranteed equal rights for women, universal adult suffrage, and freedom to form unions.
 - ✓ **Jinnah’s Fourteen Points (1929):** The Muslim League, led by Muhammad Ali Jinnah, ultimately rejected the Nehru Report because they felt it did not secure adequate political safeguards for Muslims. In response, Jinnah drafted his famous “Fourteen Points” as a counter-proposal, demanding a federal constitution with residual powers vested in the provinces and one-third Muslim representation in the Central Legislature.

Q 67. Ans. C

Explanation:

- The **Government of India Act 1935** marked a major milestone by abolishing the system of 'dyarchy' in the provinces (which had been introduced earlier by the Government of India Act 1919) and replacing it with 'Provincial Autonomy'. Under this new system, the provinces were allowed to act as autonomous units of administration within their defined spheres, and ministers responsible to the provincial legislatures were put in charge of provincial administration.
- **Indian Councils Act 1909 (Morley-Minto Reforms):** Known for introducing separate electorates for Muslims.
- **Government of India Act 1919 (Montagu-Chelmsford Reforms):** Introduced dyarchy (division of subjects into 'reserved' and 'transferred') at the provincial level, not its abolition.
- **Indian Independence Act 1947:** Partitioned British India into two independent dominions (India and Pakistan).
- **Additional Facts:**
 - ✓ **Dyarchy at the Centre:** While the Act abolished dyarchy in the provinces, it provided for the adoption of dyarchy at the Federal (Central) level. However, this provision never came into operation.
 - ✓ **Division of Powers:** It introduced a three-fold division of legislative powers between the Centre and the provinces: the Federal List, the Provincial List, and the Concurrent List. Residuary powers were vested in the Viceroy.
 - ✓ **All-India Federation:** The Act proposed the creation of an All-India Federation consisting of provinces and princely states. However, this federation never materialised because the princely states largely refused to join.
 - ✓ **New Institutions:** The Act mandated the establishment of several key institutions that form the bedrock of modern India, including the **Reserve Bank of India (RBI)** to control currency and credit, and the **Federal Court** (which was set up in 1937 and later became the Supreme Court of India).
 - ✓ **Bicameralism:** It introduced a bicameral legislature (an upper and lower house) in six out of the eleven provinces: Bengal, Bombay, Madras, Bihar, Assam, and the United Provinces.

Q 68. Ans. C

Explanation:

- **Statement 1 is correct:** The Nawab of Junagadh (a Muslim ruler over a Hindu-majority state) initially announced accession to Pakistan. Following a mass uprising and the formation of a provisional government (Arzi Hukumat) by the citizens, the Nawab fled to Karachi. India intervened, and a plebiscite was held in February 1948, where over 99% of the population voted to join India.
- **Statement 2 is incorrect:** Hyderabad did not integrate peacefully. The Nizam, Mir Osman Ali Khan, wanted to remain independent and his paramilitary force, the Razakars, unleashed violence on the population. India had to launch a military offensive code-named **Operation Polo** (often termed a "police action") in September 1948 to force the Nizam's surrender and integrate the state.
- **Statement 3 is correct:** To convince the sceptical princes to join the Indian Union before independence, Sardar Vallabhbhai Patel and V.P. Menon drafted a basic Instrument of Accession. It required the princes to surrender jurisdiction to the central government over only three core subjects: Defence, External Affairs, and Communications, leaving their internal autonomy temporarily intact.
- **Additional Facts:**
 - ✓ **The States Department:** In July 1947, the States Department was created under Sardar Vallabhbhai Patel, with V.P. Menon as its Secretary, specifically tasked with negotiating the accession of over 560 princely states.
 - ✓ **Kashmir's Accession:** Maharaja Hari Singh of Jammu & Kashmir initially opted for a "Standstill Agreement" to remain independent. He signed the Instrument of Accession to India on October 26, 1947, only after Pashtun tribal militias, backed by the Pakistani army, invaded the state.
 - ✓ **Privy Purses:** To facilitate full political integration (moving beyond the initial three subjects), the Government of India offered the former rulers a "Privy Purse", a guaranteed, tax-free annual pension based on the revenue of their former states. This system was later abolished in 1971 by Indira Gandhi via the 26th Constitutional Amendment.
 - ✓ **Travancore and Jodhpur:** Travancore (in Kerala) was one of the first states to declare

its intention to remain independent, driven by its Dewan, Sir C.P. Ramaswami Iyer (who backed down after an assassination attempt). The young Maharaja of Jodhpur, Hanwant Singh, secretly negotiated with Muhammad Ali Jinnah to join Pakistan but was ultimately persuaded by Lord Mountbatten and Sardar Patel to join India.

Q 69. Ans. D

Explanation:

- **Pair 1 is correct:** Lord Macaulay's famous "Minute on Indian Education" was delivered in 1835. It strongly advocated for Western education and made English the official medium of instruction for higher education, leading Governor-General Lord William Bentinck to pass the English Education Act of 1835.
- **Pair 2 is correct:** Sir Charles Wood's Despatch of 1854 is considered the "Magna Carta of English Education in India." It recommended a comprehensive system from primary schools to universities and advocated for vernacular languages at the primary level, Anglo-vernacular at the high school level, and English for higher studies.
- **Pair 3 is correct:** The Hunter Education Commission was appointed by Lord Ripon in 1882 (chaired by W.W. Hunter) to review the progress made since Wood's Despatch. It primarily focused on the expansion and improvement of primary and secondary education, recommending the transfer of primary education control to newly formed local district and municipal boards.
- **Pair 4 is correct:** The Raleigh Commission was appointed by Lord Curzon in 1902 under Sir Thomas Raleigh to investigate the conditions and prospects of universities in India. Its recommendations directly led to the controversial Indian Universities Act of 1904, which increased strict government control over universities.
- **Additional Facts:**
 - ✓ **Downward Filtration Theory:** This was the guiding philosophy prior to Wood's Despatch (heavily backed by Macaulay). The British believed they only needed to educate a small, elite class of Indians in English, who would then act as interpreters and filter down Western knowledge to the masses. Wood's Despatch officially repudiated this theory in favour of state-sponsored mass education.
 - ✓ **First Universities:** As a direct result of Wood's Despatch, the first three modern universities

in India, the University of Calcutta, the University of Bombay, and the University of Madras, were established in 1857 (the same year as the great revolt).

- ✓ **Sadler Commission (1917):** Also known as the Calcutta University Commission, it recommended a 12-year school course (the precursor to the 10+2 system) and the separation of secondary education from university control.
- ✓ **Wardha Scheme of Basic Education (1937):** Rooted in Mahatma Gandhi's philosophy of Nai Talim, this scheme (formulated by a committee headed by Dr. Zakir Husain) advocated for free and compulsory education in the mother tongue for 7 years, heavily centred around manual productive work (like spinning or weaving).

Q 70. Ans. A

Explanation:

- **Aligarh Movement (a - 4):** Founded by Sir Syed Ahmed Khan in the late 19th century. Its base was the Mohammedan Anglo-Oriental College (which later became Aligarh Muslim University), and its primary objective was to impart modern Western and scientific education to Indian Muslims while retaining Islamic values.
- **Singh Sabha Movement (b - 3):** Founded in Amritsar, Punjab in 1873. Its main goals were to restore the purity of Sikhism, counter the proselytising activities of Christian missionaries and the Arya Samaj, and promote modern education among Sikhs.
- **Rahnumai Mazdayasnan Sabha (c - 2):** Founded in 1851 by Naoroji Furdonji, Dadabhai Naoroji, S.S. Bengalee, and others. It was a reform movement within the Parsi (Zoroastrian) community aimed at removing orthodox customs, modernising Parsi social structures, and improving the status of women.
- **Self-Respect Movement (d - 1):** Launched in 1925 by E.V. Ramasamy Naicker (affectionately known as Periyar) in South India (Tamil Nadu). It was a radical socio-political movement aimed at dismantling the caste system, fighting Brahminical dominance, and promoting equal rights for marginalised castes and women without invoking religion.
- **Additional Facts:**
 - ✓ **Rast Goftar:** To spread their message of reform, the Rahnumai Mazdayasnan Sabha

heavily utilised the Gujarati newspaper Rast Goftar (The Truth Teller), which was edited by Dadabhai Naoroji.

- ✓ **Tahdhib-ul-Akhlaq:** Sir Syed Ahmed Khan published this influential journal (meaning “Social Reformer”) to awaken the Muslim masses to the need for rational thinking, social reform, and scientific education.
- ✓ **Khalsa College:** The Singh Sabha movement spearheaded a massive educational drive in the Punjab region, which culminated in the establishment of the prestigious Khalsa College in Amritsar in 1892.
- ✓ **Vaikom Satyagraha (1924–25):** Just before formally launching the Self-Respect Movement, Periyar played a pivotal role in the Vaikom Satyagraha in Kerala, a major agitation demanding the right of lower castes (avarnas) to use the public roads surrounding the Vaikom Mahadeva Temple.

Q 71. Ans. D

Explanation:

- **Pair 1 is correct:** The Bengal Sati Regulation of 1829 (Regulation XVII) legally banned the practice of Sati in all jurisdictions of British India. It was enacted by the Governor-General Lord William Bentinck, heavily driven by the tireless campaigning of Raja Ram Mohan Roy.
- **Pair 2 is correct:** The Indian Slavery Act of 1843 (Act V of 1843) officially abolished slavery in British India. It was passed under the Governor-Generalship of Lord Ellenborough.
- **Pair 3 is correct:** The Hindu Widows’ Remarriage Act of 1856 legalised the remarriage of Hindu widows in all jurisdictions of India. It was drafted by Lord Dalhousie and passed by Lord Canning before the 1857 revolt, largely due to the relentless efforts of Ishwar Chandra Vidyasagar.
- **Pair 4 is correct:** The Age of Consent Act of 1891 raised the age of consent for sexual intercourse for all girls (married or unmarried) from 10 to 12 years. It was enacted under Lord Lansdowne, spearheaded by the Parsi social reformer Behramji Malabari.
- **Additional Facts:**
 - ✓ **Child Marriage Restraint Act (1929):** Popularly known as the **Sarda Act** (after its sponsor Harbilas Sarda), this landmark legislation fixed the minimum marriageable age at 14 years for girls and 18 years for boys across all communities in British India.

- ✓ **Native Marriage Act (1872):** Also known as the Civil Marriage Act III of 1872, it was passed primarily due to the efforts of Keshub Chandra Sen (of the Brahmo Samaj). It legalised inter-caste marriages, prohibited polygamy, and set the minimum age of marriage at 14 for girls and 18 for boys (though it initially only applied to those who declared they did not belong to recognised orthodox faiths).
- ✓ **Female Infanticide Prevention Act (1870):** To tackle the rampant practice of female infanticide (especially prevalent in parts of North-Western Provinces, Punjab, and Rajputana), this act mandated the compulsory registration of births and allowed for strict police surveillance in districts where the sex ratio was highly skewed.

Q 72. Ans. B

Explanation:

- The personality being described is **Sachindranath Sanyal**. His revolutionary career began in Benaras (Varanasi), where he was deeply influenced by Mokhodacharan Samadhyay, a veteran revolutionary from Bengal who took over the nationalist newspaper Sandhya after the death of its founder, Brahmabandhab Upadhyay.
- Sanyal went on to become one of the most prominent revolutionary leaders in North India. In 1924, he co-founded the **Hindustan Republican Association (HRA)** in Kanpur alongside Ram Prasad Bismil, Jogesh Chandra Chatterjee, and others.
- **Additional Facts:**
 - ✓ **Bandi Jiwan:** While serving his sentence at the Cellular Jail in the Andaman Islands, he wrote his famous autobiography, *Bandi Jiwan* (A Life of Captivity). This book became the essential “textbook” and major source of inspiration for a whole generation of younger revolutionaries, including Bhagat Singh and Chandrashekhar Azad.
 - ✓ **The Revolutionary:** Sanyal drafted the famous manifesto of the HRA titled *The Revolutionary*, which was distributed across major cities in North India on New Year’s Day in 1925, calling for a federal republic of the United States of India achieved through an armed rebellion.
 - ✓ **Conspiracies:** He was uniquely involved in two major revolutionary conspiracies of the era. He was a close associate of Rash Behari Bose in the Delhi-Lahore Conspiracy (the

1912 assassination attempt on Viceroy Lord Hardinge) and was later given a life sentence for his role in the Kakori Train Robbery (1925).

Q 73. Ans. A

Explanation:

- **Satyashodhak Movement (1873):** Founded by Jyotirao Phule in Pune, Maharashtra. The Satyashodhak Samaj (Truth Seekers' Society) aimed to liberate the lower castes and untouchables from the exploitation and oppression of Brahmins.
- **Aravippuram Movement (1888):** Initiated by Sri Narayana Guru in Kerala. He defied the traditional Brahmin monopoly on religious rites by consecrating a Shiva idol (the Aravippuram Shiva) himself, sparking a massive social reform movement among the Ezhavas.
- **Justice Party Movement (1916):** Formally known as the South Indian Liberal Federation (SILF), it was founded by Dr. C. Natesa Mudaliar, T. M. Nair, and P. Theagaraya Chetty in Madras. It aimed to secure jobs and representation for non-Brahmins in the legislature.
- **Self-Respect Movement (1925):** Founded by E.V. Ramasamy Naicker (Periyar) in Tamil Nadu. It was a radical offshoot that aimed to dismantle the caste system entirely, rejecting Brahminical religion and promoting rationalism and equal rights.
- **Additional Facts:**
 - ✓ **Gulamgiri:** Jyotirao Phule wrote the influential book *Gulamgiri* (Slavery) in 1873, dedicating it to the American anti-slavery movement, thereby linking the plight of India's lower castes with that of enslaved Black Americans.
 - ✓ **SNDP Yogam:** Arising from the Aravippuram movement, the Sree Narayana Dharma Paripalana (SNDP) Yogam was founded in 1903 to carry forward Narayana Guru's vision, famously encapsulated in his slogan: "One Caste, One Religion, One God for Mankind."
 - ✓ **Dravidar Kazhagam:** In 1944, the Justice Party and the Self-Respect Movement merged under Periyar's leadership to form the Dravidar Kazhagam (DK), which later gave birth to major political parties like the DMK and AIADMK in Tamil Nadu.

Q 74. Ans. A

Explanation:

- **Somaprakash (1858):** Started by Ishwar Chandra Vidyasagar and edited by Dwarkanath

Vidyabhusan. It was a pioneering Bengali weekly that boldly championed the cause of the indigo peasants and criticised the British administration.

- **National Paper (1865):** Founded by Nabagopal Mitra in Calcutta. It was closely associated with the Hindu Mela (originally known as the Jatiya Mela or National Gathering) and aimed to promote nationalism and self-reliance.
- **Amrita Bazar Patrika (February 1868):** Founded by brothers Sisir Kumar Ghosh and Motilal Ghosh in Jessore (now in Bangladesh) before moving to Calcutta. It originally started as a Bengali weekly.
- **Madras Mail (December 1868):** Founded by Charles Lawson and Henry Cornish. It was the first evening newspaper in India, started as a joint-stock company.
- **Additional Facts:**
 - ✓ When Viceroy Lord Lytton passed the draconian Vernacular Press Act in 1878 to suppress local language newspapers (specifically targeting papers like *Somaprakash* and *Amrita Bazar Patrika*), *Amrita Bazar Patrika* famously bypassed the law by transforming entirely into an English-language newspaper overnight.
 - ✓ **The First Newspaper:** The very first newspaper in India was the *Bengal Gazette* (also known as the *Calcutta General Advertiser*), started by the Irishman **James Augustus Hicky** in 1780. It was heavily critical of Governor-General Warren Hastings and was quickly seized.
 - ✓ **Hindu Patriot:** Founded in 1853 by Madhusudan Ray and prominently edited by Harish Chandra Mukherjee, this English weekly became the powerful voice of the ryots (peasants) during the Indigo Revolt (1859-60).

Q 75. Ans. B

Explanation:

- The slogan "Jai Hind" (Victory to India) was coined by **Zain-ul Abideen Hasan** (also known as Abid Hasan Safrani), a close aide and commander in the Indian National Army (INA). Subhas Chandra Bose tasked him with finding a common, secular military greeting that could unite Indians across different castes and religions. Hasan originally suggested "Hello", which Bose rejected, before coming up with "Jai Hind."
- While Subhas Chandra Bose did not coin the phrase, he enthusiastically adopted it as the official greeting and battle cry of the INA, ultimately popularising it across the nation.
- **Additional Facts:**

- ✓ **Abid Hasan Safrani:** He was the only Indian who accompanied Subhas Chandra Bose on his perilous 93-day submarine journey from Germany to Madagascar (and eventually to Japan) during World War II. He added "Safrani" to his name as a symbol of communal harmony, honouring the saffron colour of the Hindus while being a Muslim himself.
- ✓ **Subhas Chandra Bose:** While he didn't coin "Jai Hind", Bose is credited with giving the famous slogans "Dilli Chalo" (Onward to Delhi) and "Give me blood, and I shall give you freedom!" to motivate the INA soldiers.
- ✓ **Inquilab Zindabad:** Often associated closely with Bhagat Singh who popularised it, this famous slogan (meaning "Long Live the Revolution") was actually coined by the Urdu poet and freedom fighter **Maulana Hasrat Mohani** in 1921.

Q 76. Ans. D

Explanation:

- **Milap (b - 2):** A prominent Urdu daily founded by Mahashay Krishna in Lahore in **1923**.
- **Bahishkrit Bharat (c - 1):** A Marathi fortnightly launched by Dr. B.R. Ambedkar in April **1927** to champion the rights of the depressed classes.
- **Kirti (d - 4):** A Punjabi communist journal started by Bhai Santokh Singh in **1926** to organise workers and peasants.
- **The Hindustan Times (a - 3):** Founded by Sunder Singh Lyallpuri and inaugurated by Mahatma Gandhi in 1924.
- **Additional Facts:**
 - ✓ **Dr. Ambedkar's Press:** Before Bahishkrit Bharat, Dr. Ambedkar started his first publication, the Marathi fortnightly Mooknayak (Leader of the Silent), in 1920 with the financial assistance of Chhatrapati Shahu Maharaj of Kolhapur.
 - ✓ **Kirti Kisan Party:** The Kirti journal was so influential that it led to the formation of the Kirti Kisan Party (Workers and Peasants Party) in Punjab, blending Marxist ideology with the revolutionary spirit of the Ghadar movement. Bhagat Singh actively wrote for this journal under the pseudonym "Vidrohi".
 - ✓ **Hindustan Times First Editor:** The eminent historian, diplomat, and scholar K.M. Panikkar served as the first editor of The Hindustan Times, which was started primarily to voice the grievances of the Sikh Akali Dal.

Q 77. Ans. C

Explanation:

- The Canadian government did not allow all the passengers to land. Despite a two-month legal battle and negotiations, only 24 passengers were admitted to Canada. The remaining 352 passengers were forced to remain on the ship and were escorted out of Canadian waters by the military, forced to return to India.
- **Additional Facts:**
 - ✓ **Gurdit Singh:** The chartered Japanese steamship Komagata Maru was hired by Gurdit Singh, a Singapore-based Sikh businessman, to challenge the discriminatory Canadian exclusion laws.
 - ✓ When the ship was forced to return and docked at Budge Budge near Calcutta in September 1914, the British police treated the passengers as dangerous political agitators. They attempted to force them onto a train to Punjab. A riot broke out, and the British police opened fire, killing 20 passengers and arresting many others.
 - ✓ This Canadian law (passed in 1908) was a deliberate loophole designed to keep Indians out. It required immigrants to travel by a continuous journey from their home country. Since there were no direct shipping lines from India to Canada at the time, it served as an effective, racially motivated ban without explicitly naming Indians.

Q 78. Ans. D

Explanation:

- **Pair 2 is correct:** Guru Amar Das (the 3rd Sikh Guru) introduced the **Manji system**. Because the Sikh community had grown significantly and he was of advanced age, he divided the Sikh congregation into 22 dioceses or Manjis (seats of authority), each headed by a devoted Sikh to spread the teachings of Guru Nanak.
- **Pair 3 is correct:** Guru Ram Das (the 4th Sikh Guru) initiated the **Masand system**. Masands were official representatives who were deputed to preach the Sikh faith and collect offerings (funds) from the followers, which were primarily used to finance the construction of the holy city of Amritsar and the Sarovar (sacred pool).
- **Pair 4 is correct:** Guru Gobind Singh (the 10th Sikh Guru) established the **Khalsa Panth** on Baisakhi day

in 1699 at Anandpur Sahib, transforming the Sikhs into a martial community (the “Saint-Soldiers”) and initiating the Panj Pyare (the Five Beloved Ones).

● **Additional Facts:**

- ✓ **Dasvandh:** While Guru Ram Das started the Masand system for collection, it was Guru Arjan Dev (the 5th Guru) who formalised the practice of Dasvandh, requiring Sikhs to donate exactly one-tenth of their income to the Guru’s treasury for community welfare and the construction of the Harmandir Sahib (Golden Temple).
- ✓ **Abolition of Masands:** Over time, the Masands became highly corrupt, extortionist, and power-hungry. Recognising this rot, Guru Gobind Singh completely abolished the Masand system when he created the Khalsa, establishing a direct relationship between the Guru and the Sikh congregation.
- ✓ **Miri and Piri:** Following the execution of Guru Arjan Dev by the Mughal Emperor Jahangir, his son Guru Hargobind (the 6th Guru) militarised the Sikh community. He introduced the concept of Miri (temporal power) and Piri (spiritual authority), symbolised by him wearing two swords. He also constructed the Akal Takht opposite the Harmandir Sahib.
- ✓ **Adi Granth:** The first official edition of the Sikh scripture, the Adi Granth, was compiled by Guru Arjan Dev in 1604. It was later expanded by Guru Gobind Singh to include the hymns of Guru Tegh Bahadur, resulting in the final version, the **Guru Granth Sahib**, which was declared the eternal, living Guru of the Sikhs in 1708.

Q 79. Ans. A

Explanation:

- Sher Shah Suri restored and vastly expanded the ancient trans-continental route in the 16th century, renaming it **Sadak-i-Azam** (The Great Road). It stretched over 2,500 kilometres, connecting Sonargaon (in present-day Bangladesh) to Peshawar and Kabul (in present-day Pakistan and Afghanistan). During the British era, this monumental highway was renamed the **Grand Trunk Road** (often abbreviated as the GT Road).
- **Additional Facts:**
 - ✓ **Ancient Origins (Uttarapatha):** Long before Sher Shah, the foundation of this route was laid by the Mauryan Emperor Chandragupta

Maurya in the 3rd century BCE. It was known as the Uttarapatha, stretching from the mouth of the Ganges to the north-western frontier of the empire.

- ✓ **Sarais (Rest Houses):** To facilitate trade, travel, and a rapid postal service, Sher Shah built a network of over 1,700 sarais (inns) at regular intervals along his roads. These inns provided separate quarters for Hindus and Muslims, along with food and water for travellers and their horses.
- ✓ **Kos Minars:** Successive Mughal emperors (especially Akbar and Jahangir) maintained and expanded upon Sher Shah’s work, erecting solid brick pillars known as Kos Minars (milestones) every two kos (roughly 3.2 kilometres) to help travellers gauge distances.
- ✓ **Administrative Legacy:** Besides the road network, Sher Shah Suri is famous for introducing the silver coin known as the **Rupiya** (the precursor to the modern Rupee) and establishing a highly efficient land revenue system (the Zabt system), which was later adopted and perfected by Akbar’s finance minister, Todar Mal.

Q 80. Ans. A

Explanation:

- **Statement 1 is correct:** Sher Shah completely overhauled the chaotic currency system of the Delhi Sultanate. He introduced a standardised silver coin weighing 178 grains called the Rupiya, which served as the basis for the modern Indian Rupee. He also issued copper coins known as Dam and gold coins known as Mohur.
- **Statement 2 is correct:** To protect farmers from arbitrary extortion by tax collectors, he introduced the Patta and Qabuliyat system. The Patta was a title deed given to the peasant specifying the area of land they cultivated and the exact revenue amount they owed. The Qabuliyat was the corresponding deed of agreement signed by the peasant accepting these terms.
- **Statement 3 is incorrect:** The state’s share of land revenue under Sher Shah was generally fixed at **one-third (1/3)** of the average agricultural produce, not half. To calculate this, land was surveyed and classified into three categories (good, middle, and bad) based on fertility. (It was Alauddin Khalji who famously raised the land revenue to 50%).
- **Additional Facts:**

- ✓ **Standardised Measurement:** His revenue assessment was based on the precise measurement of land using a standardised measuring scale called the Sikandari Gaj (which was roughly 32 inches, originally introduced by Sikandar Lodi).
- ✓ **Raja Todar Mal's Inspiration:** Sher Shah's highly efficient land revenue system laid the direct foundation for the later Mughal administration. Raja Todar Mal, who would later become Emperor Akbar's legendary finance minister and formulate the Zabti or Dahsala system, began his administrative career serving under Sher Shah Suri.
- ✓ **Military Discipline:** To maintain a strong and loyal standing army, Sher Shah revived the strict military practices of Dagh (the branding of horses) and Chehra (maintaining descriptive rolls of soldiers) to prevent corruption and the substitution of inferior horses or untrained troops during musters.

Q 81. Ans. C

Explanation:

- The passage accurately describes the **Phillips Curve**, an economic concept developed by New Zealand economist A.W. Phillips in 1958.
- The theory posits an inverse, stable relationship between unemployment and inflation in the short term. When unemployment is high, inflation is low, and vice versa.
- Historically, policymakers used this curve to justify expansionary fiscal or monetary policies (like increased government spending or lowering interest rates) to boost aggregate demand and lower unemployment, accepting a slight rise in inflation as the inevitable cost.
- In the 1970s, many economies experienced stagflation, a simultaneous spike in both inflation and unemployment, which completely contradicted the original Phillips Curve. This led economists like Milton Friedman to argue that the trade-off only exists in the short run. In the long run, the curve is vertical; unemployment naturally returns to its structural rate (NAIRU) regardless of inflation.
- **Additional Facts:**
 - ✓ **Ricardian Equivalence:** An economic theory (originating from David Ricardo and expanded by Robert Barro) suggesting that consumer spending remains unchanged whether a government finances its spending through

current taxes or by issuing debt. Consumers rationally anticipate that government debt today means higher taxes tomorrow, so they save their excess money rather than spending it.

- ✓ **Laffer Curve:** Illustrates the relationship between tax rates and tax revenue. It suggests there is an optimal tax rate that maximizes government revenue; taxing beyond this point actually decreases total revenue because it heavily discourages work, production, and investment.
- ✓ **Kuznets Curve:** Hypothesizes that as a developing economy grows, economic inequality first increases, peaks, and then eventually decreases (forming an inverted U-shape) as the nation industrializes and establishes a welfare state.
- ✓ **Nash Equilibrium:** (Related to the "Nashian" option) A foundational concept in game theory where optimal outcome is reached when no player has an incentive to deviate from their chosen strategy, assuming all other players keep their strategies constant.

Q 82. Ans. C

Explanation:

- **Statement 1 is correct:** The World Trade Organisation (WTO) was established on 1 January 1995, replacing the General Agreement on Tariffs and Trade (GATT), which had governed international trade since 1948. The WTO emerged from the Uruguay Round of negotiations (1986–1994) and was formally created through the Marrakesh Agreement. Unlike GATT, which was merely a provisional multilateral agreement, the WTO is a full-fledged international organisation with a permanent institutional structure and a binding dispute settlement mechanism.
- **Statement 2 is incorrect:** Maintaining exchange rate stability and overseeing the international monetary system is the core mandate of the International Monetary Fund (IMF), not the WTO. The WTO's role is instead focused on administering trade agreements, providing a forum for trade negotiations, settling trade disputes between member countries, monitoring national trade policies, and providing technical assistance/training to developing countries, it has no jurisdiction over currency or exchange rate matters.
- **Statement 3 is correct:** The Trade Policy Review

Body (TPRB) is a crucial mechanism of the WTO. The General Council (the WTO's highest-level decision-making body in Geneva) meets as the TPRB to conduct regular, peer-reviewed assessments of the national trade policies of its member countries to ensure transparency.

● **Additional Facts:**

- ✓ **Marrakesh Agreement:** The WTO was created as a result of the Marrakesh Agreement, which was signed in April 1994 at the conclusion of the grueling, 8-year-long Uruguay Round of GATT negotiations.
- ✓ **Dispute Settlement Body (DSB):** Arguably the most powerful arm of the WTO is its dispute settlement mechanism. If a member believes another member is violating trade agreements, they can take the issue to the DSB, which has the authority to authorize retaliatory trade sanctions against the offending country.
- ✓ **Most-Favored-Nation (MFN) Principle:** A foundational pillar of the WTO is the MFN principle, which dictates that countries cannot normally discriminate between their trading partners. If a country grants someone a special favor (such as a lower customs duty rate for one of their products), they must do the same for all other WTO members.
- ✓ **Headquarters & Leadership:** The WTO is headquartered in Geneva, Switzerland. Unlike the World Bank or IMF, the WTO is not technically a specialised agency of the United Nations (though it maintains strong ties with the UN).

Q 83. Ans. B

Explanation:

- The Fair and Remunerative Price (FRP) is the minimum price that sugar mills are legally bound to pay to farmers for **sugarcane**. It replaced the earlier Statutory Minimum Price (SMP) system in 2009 to ensure that farmers receive a guaranteed, fair margin over their cost of cultivation. It is governed by the Sugarcane (Control) Order, 1966.
- **Wheat & Jute:** These are covered under the Minimum Support Price (MSP) system, not the FRP.
- **Tobacco:** Regulated by the Tobacco Board, and highly graded varieties are often subject to auction systems, though sometimes covered by specific MSP interventions if prices crash.
- **Additional Facts:**

✓ The FRP (as well as MSP for other crops) is recommended by the **Commission for Agricultural Costs and Prices (CACP)**. It is then officially approved and announced by the **Cabinet Committee on Economic Affairs (CCEA)**, which is chaired by the Prime Minister.

✓ **FRP vs. MSP Mechanism:** Under the MSP system, if market prices fall below the declared price, the government steps in to procure the crops (like wheat or paddy) directly from the farmers. Under the FRP system, the government does not buy the sugarcane; instead, it legally mandates that private and cooperative **sugar mills** must pay the farmers the FRP.

✓ **State Advised Price (SAP):** Several major sugarcane-producing states (such as Uttar Pradesh, Punjab, and Haryana) announce their own pricing called the State Advised Price (SAP). The SAP is almost always higher than the central government's FRP, and sugar mills in these states are legally bound to pay the higher SAP to the farmers.

✓ **Sugar Recovery Rate:** The FRP is usually linked to a basic "sugar recovery rate" (the amount of sugar extracted from a given weight of cane). If a farmer's cane yields a higher recovery rate than the base level, they receive a premium over the standard FRP.

Q 84. Ans. D

Explanation:

- The proceeds from the disinvestment of Central Public Sector Enterprises (CPSEs) are channeled into the **National Investment Fund (NIF)**.
- The Government of India constituted the NIF in November 2005. Initially, the corpus of the fund was managed by public sector mutual funds, and the returns were used to finance select social sector schemes and capital investment in profitable CPSEs.
- **Additional Facts:**
 - ✓ **Restructuring of NIF (2013):** The government restructured the NIF so that the entire disinvestment proceeds are now credited to the existing 'Public Account' under the NIF head. These funds can now be used for a broader range of purposes, including the recapitalization of Public Sector Banks (PSBs) and investment in entities like NABARD and RRBs.

- ✓ **DIPAM:** The nodal agency responsible for the management of government equity in CPSEs and the execution of the disinvestment policy is the **Department of Investment and Public Asset Management (DIPAM)**, which operates under the Ministry of Finance.
- ✓ **Strategic vs. Minority Disinvestment:** Minority disinvestment involves the government selling a portion of its shares while retaining majority control (over 51%). Strategic disinvestment (or strategic sale) involves selling a substantial portion of the government's shareholding (up to 100%) along with the transfer of management control to a private entity (e.g., the privatization of Air India).
- ✓ **NIIF:** The National Investment and Infrastructure Fund (NIIF) is India's first sovereign wealth fund, established in 2015 to attract foreign and domestic capital into the country's infrastructure sector, not a repository for disinvestment proceeds.

Q 85. Ans. A

Explanation:

- **Statement 1 is correct:** The Securities Appellate Tribunal (SAT) is a statutory and independent body established under the provisions of the Securities and Exchange Board of India (SEBI) Act, 1992. Its primary function is to hear and dispose of appeals against orders passed by SEBI or its adjudicating officers. (Note: SAT also hears appeals against orders passed by PFRDA and IRDAI).
- **Statement 2 is correct:** SAT consists of a Presiding Officer and two other Members (one Judicial Member and one Technical Member). The Presiding Officer is appointed by the Central Government in consultation with the Chief Justice of India or his nominee.
- **Statement 3 is incorrect:** The SEBI Chairman does not need the approval of SAT for search and seizure operations. Under the SEBI Act (amended in 2014 to grant SEBI more teeth), if the investigating authority has reasonable grounds to believe that documents will be destroyed or hidden, the SEBI Chairman can authorize search and seizure, but this requires an order/approval from a **Magistrate** or a Judge of a designated court in Mumbai, not the Securities Appellate Tribunal.
- **Additional Facts:**
 - ✓ **Civil Court Powers:** SAT operates with the same powers as vested in a civil court under

the Code of Civil Procedure, 1908 (such as summoning witnesses, requiring document production, and receiving evidence on affidavits).

- ✓ **Supreme Court Appeals:** If a person or entity is aggrieved by a decision or order made by SAT, they can file an appeal directly to the **Supreme Court of India** (within 60 days), but only on questions of law.
- ✓ **SEBI's Structure:** SEBI functions as a quasi-legislative, quasi-judicial, and quasi-executive body. It drafts regulations, conducts inquiries, passes rulings, and imposes penalties. It was initially formed in 1988 as a non-statutory body and was granted statutory status on January 30, 1992.

Q 86. Ans. A

Explanation:

- **Statement 1 is correct:** The Stand-Up India Scheme facilitates bank loans ranging from **₹10 lakh to ₹1 crore** to eligible borrowers.
- **Statement 2 is correct:** The core objective of the scheme is to promote entrepreneurship among underserved demographics. It mandates every scheduled commercial bank branch to provide this loan to at least one **Scheduled Caste (SC) or Scheduled Tribe (ST)** borrower and at least one **woman** borrower for setting up a **greenfield enterprise**.
- **Statement 3 is incorrect:** Providing collateral-free working capital loans to street vendors is the objective of the **PM SVANidhi** (Pradhan Mantri Street Vendor's AtmaNirbhar Nidhi) scheme, not the Stand-Up India Scheme.
- **Additional Facts:**
 - ✓ **Greenfield Meaning:** In the context of this scheme, a "greenfield enterprise" strictly means the very first time the beneficiary is setting up a venture in the manufacturing, services, agri-allied activities, or trading sector. It does not apply to the expansion of an existing business.
 - ✓ **Non-Individual Enterprises:** In the case of non-individual enterprises (like private limited companies or partnerships), at least 51% of the shareholding and controlling stake must be held by either an SC/ST or a woman entrepreneur.
 - ✓ **Margin Money Reduction:** Originally, the scheme required borrowers to bring in 25% of

the project cost as margin money. In 2021, the government amended the scheme to reduce the margin money requirement to 15% to make it more accessible.

- ✓ **SIDBI's Role:** The scheme is anchored by the Department of Financial Services (DFS), Ministry of Finance. The Small Industries Development Bank of India (SIDBI) acts as the refinancing agency for the loans disbursed by commercial banks under this initiative

Q 87. Ans. B

Explanation:

- **Statement 1 is correct:** India's first Central Bank Digital Currency (CBDC)-based Public Distribution System (PDS) pilot was recently launched in Gandhinagar, Gujarat, in February 2026. The system uses digital tokens to securely transfer PDS subsidies to beneficiaries.
- **Statement 2 is correct:** A major advantage of CBDC is "programmability." This allows the Reserve Bank of India to program digital food coupons so that they are "purpose-bound"—meaning the funds can only be redeemed for specific food grains at authorized Fair Price Shops, preventing the diversion of subsidies.
- **Statement 3 is incorrect:** The digital rupee (₹) is a centralized fiat currency issued and regulated directly by the **Reserve Bank of India (RBI)**, not the National Payments Corporation of India (NPCI). Unlike cryptocurrencies, it is not a decentralized virtual asset.
- **Additional Facts:**
 - ✓ **Annapurti ATMs:** Along with the CBDC-based PDS, automated 'Annapurti Grain ATMs' have been deployed in Gujarat. These machines can accurately dispense up to 25 kilograms of grain in just 35 seconds, drastically reducing human error and middleman corruption.
 - ✓ **Financial Inclusion:** The system does not strictly require a traditional bank account or a smartphone. Beneficiaries without smartphones can authenticate and redeem their programmable tokens using Aadhaar-based OTPs at Fair Price Shops.
 - ✓ **CBDC vs. UPI:** While UPI (overseen by NPCI) is a payment interface used to transfer money between existing bank accounts, CBDC is the digital form of the sovereign currency itself, acting as a direct liability of the RBI.

Q 88. Ans. B

Explanation:

- **Statement 1 is correct:** Catastrophe (Cat) bonds are high-yield debt instruments designed to raise money for companies in the insurance industry in the event of a natural disaster. If the specified "trigger event" (e.g., a hurricane of Category 5 hitting Florida, or an earthquake of magnitude 7.0 in Tokyo) occurs, the investors lose some or all of their principal, which is transferred to the sponsor to pay out insurance claims. If the disaster does not occur, investors get their principal back plus interest.
- **Statement 2 is incorrect:** Cat bonds carry significant risk (the risk of a total loss of principal due to an unpredictable natural disaster). To compensate investors for taking on this extreme risk, they offer **substantially higher** interest rates compared to regular corporate or government bonds.
- **Statement 3 is incorrect:** One of the biggest selling points of Cat bonds for institutional investors is that they are **uncorrelated** with traditional financial markets or global economic cycles. A stock market crash, inflation, or a recession has no impact on whether an earthquake or hurricane strikes. Thus, they provide excellent portfolio diversification.
- **Statement 4 is correct:** They are primarily issued by insurance and reinsurance companies (and sometimes governments or large corporations) to transfer peak natural disaster risks directly to the global capital markets, protecting themselves from insolvency in the wake of a mega-disaster.
- **Additional Facts:**
 - ✓ **Origins:** The Cat bond market was born in the mid-1990s in the aftermath of **Hurricane Andrew (1992)**, which caused such massive devastation in Florida that several insurance companies went bankrupt. The industry realized it needed to tap into the broader capital markets to handle such extreme events.
 - ✓ **Parametric vs. Indemnity Triggers:** Cat bonds typically pay out based on one of two triggers. An indemnity trigger pays out based on the actual financial losses the insurer suffers. A parametric trigger pays out based purely on the physical parameters of the disaster (e.g., wind speed recorded at a specific weather station, or earthquake magnitude at a specific depth), allowing for much faster payouts since it requires no lengthy damage assessment.

- ✓ **World Bank Issuances:** The World Bank frequently issues Cat bonds on behalf of developing nations. For example, the World Bank has issued Cat bonds for countries like Mexico, Jamaica, and the Philippines to provide them with immediate, debt-free liquidity in the immediate aftermath of severe earthquakes or cyclones.

Q 89. Ans. C

Explanation:

- **Statement 1 is correct:** The Rapid Financial Instrument (RFI) is a lending facility created by the **International Monetary Fund (IMF)** to make its financial support more flexible and responsive to the diverse needs of member countries.
- **Statement 2 is incorrect:** The RFI is available to **all** member countries of the IMF. (The IMF has a separate, parallel facility called the Rapid Credit Facility, or RCF, which is exclusively designed for low-income countries and provides support on concessional terms).
- **Statement 3 is correct:** It is specifically designed to provide rapid financial assistance with limited conditionality to any member facing an urgent balance of payments (BoP) need, such as those arising from commodity price shocks, natural disasters, conflict situations, or sudden economic disruptions like the COVID-19 pandemic.
- **Statement 4 is incorrect:** Unlike standard IMF loans, the RFI is unique because it provides immediate support **without** the requirement of a full-fledged macroeconomic program or strict, long-term structural reforms (ex-post conditionality). A member country only needs to cooperate with the IMF in making efforts to solve its BoP difficulties and outline the general policies it proposes to implement.
- **Additional Facts:**
 - ✓ **Stand-By Arrangement (SBA):** This is the IMF's historical workhorse lending instrument for emerging and advanced market economies. Unlike the RFI, the SBA requires strong conditionality, meaning the borrowing country must implement strict macroeconomic reforms to receive phased disbursements over time.
 - ✓ **Extended Fund Facility (EFF):** Designed for countries facing serious medium-term balance of payments problems rooted in structural weaknesses that require time to address. It

features longer engagement and repayment periods than the SBA.

- ✓ **Special Drawing Rights (SDR):** The SDR is not a currency or a loan, but an international reserve asset created by the IMF. Its value is based on a basket of five major currencies: the US Dollar, the Euro, the Chinese Renminbi, the Japanese Yen, and the British Pound Sterling.

Q 90. Ans. A

Explanation:

- **Statement 1 is correct:** **Raajmarg Infra Investment Trust (RIIT)** is a newly established Infrastructure Investment Trust sponsored by the National Highways Authority of India (NHAI). SEBI recently approved its public issue (filed in early 2026) to monetize operational national highway assets, allowing retail and institutional investors to participate in the cash flows generated by long-term toll collections.
- **Statement 2 is incorrect:** InvITs do not exclusively fund high-risk, early-stage greenfield projects. In fact, their primary purpose is the exact opposite. They are designed to pool money from investors to invest predominantly in **completed, operational, and revenue-generating (brownfield)** infrastructure projects. This allows developers to unlock tied-up capital while providing investors with a stable, long-term yield.
- **Statement 3 is incorrect:** Infrastructure Investment Trusts are capital market instruments and are regulated directly by the **Securities and Exchange Board of India (SEBI)** under the SEBI (Infrastructure Investment Trusts) Regulations, 2014, not by the Reserve Bank of India (RBI).
- **Additional Facts:**
 - ✓ **Structure:** An InvIT operates much like a mutual fund. It consists of a Sponsor (who sets up the InvIT, like NHAI), a Trustee (who holds the assets in trust for the benefit of the unitholders), an Investment Manager (who manages the assets and investments), and a Project Manager (who undertakes the operations and maintenance of the physical assets).
 - ✓ **Dividend Distribution:** Under SEBI guidelines, InvITs are mandated to distribute a minimum of **90% of their net distributable cash flows** to their investors (unitholders) on a bi-annual basis, making them highly attractive for yield-seeking investors.

- ✓ **NHAI's Dual Trusts:** Raajmarg Infra Investment Trust (RIIT) is actually NHAI's second InvIT venture, building on the success of its first trust, the National Highways Infra Trust (NHAI InvIT), as part of the government's broader National Monetisation Pipeline (NMP) strategy.

Q 91. Ans. C

Explanation:

- **The Netherlands** does not border the Baltic Sea; its coastline lies entirely along the **North Sea**.
- **Additional Facts:**
 - ✓ **Bordering Nations:** The Baltic Sea is enclosed by nine countries: Denmark, Sweden, Finland, Russia, Estonia, Latvia, Lithuania, Poland, and Germany.
 - ✓ **Brackish Water:** The Baltic Sea is the largest body of brackish water (a mixture of saltwater and freshwater) in the world. It has very low salinity compared to the open ocean because it receives massive freshwater runoff from over 200 rivers, and its connection to the salty North Sea via the narrow Danish Straits is highly restricted.
 - ✓ **The Kiel Canal:** To bypass the long and sometimes treacherous journey around the Jutland Peninsula of Denmark, the Kiel Canal was constructed through northern Germany, directly linking the North Sea to the Baltic Sea.
 - ✓ **Exclusive Economic Zones (EEZ):** The strategic and economic importance of the Baltic Sea (including underwater infrastructure like the Nord Stream pipelines) makes its seabed a highly contested geopolitical space, heavily monitored by NATO and Russia.

Q 92. Ans. D

Explanation:

- **Conrad Discontinuity:** Named after seismologist Victor Conrad, the Conrad Discontinuity marks the boundary between the upper continental crust and the lower continental crust. Unlike the Mohorovičić Discontinuity, it is not a well-defined global boundary and is absent in several continental regions.
- **Mohorovičić Discontinuity (Moho):** Discovered in 1909 by Croatian seismologist Andrija Mohorovičić, the Mohorovičić Discontinuity is a major seismic boundary separating the Earth's crust from the mantle. The depth of the Moho varies depending

on the geological setting, being shallower beneath oceans and deeper beneath continents.

- **Repetti Discontinuity:** The Repetti Discontinuity is the seismic boundary that separates the upper mantle from the lower mantle, marking a significant change in the physical properties of mantle materials.
- **Additional facts:**
 - ✓ **Gutenberg Discontinuity:** Located at a depth of about 2,900 kilometers, this major boundary separates the solid lower mantle from the liquid outer core. Seismic S-waves completely disappear at this boundary because they cannot travel through liquid.
 - ✓ **Lehmann Discontinuity:** This marks the boundary between the liquid outer core and the solid inner core, located at a depth of roughly 5,150 kilometers. It was discovered by Danish seismologist Inge Lehmann in 1936.
 - ✓ **Seismic Waves as Tools:** We cannot physically drill past the crust (the deepest hole ever drilled, the Kola Superdeep Borehole, only reached about 12 km). All our knowledge of these interior boundaries comes from the study of seismic waves (P-waves and S-waves) generated by earthquakes, which change speed or get reflected/refracted when they hit layers of different densities.

Q 93. Ans. A

Explanation:

- **Direct sources (physical material actually obtained/observed):**
 - ✓ Rocks obtained from mining areas
 - ✓ Deep drilling projects — such as the Kola Superdeep Borehole in Russia, which reached about 12 km depth, still a tiny fraction of the Earth's ~6,370 km radius
 - ✓ Materials ejected during volcanic eruptions, which provide molten rock (magma) from within the Earth
- **Indirect sources (inferred from measurable phenomena, without physically accessing the material):**
 - ✓ Seismic waves (activity) — analysis of how P-waves and S-waves travel through and are refracted/reflected by different layers is the most important indirect source, revealing the density, state (solid/liquid), and composition of the Earth's internal layers
 - ✓ Gravitation — variations in gravitational force (gravity anomalies) at different locations reveal the distribution of mass within the Earth

- ✓ Other indirect sources include analysis of the Earth's magnetic field, and the study of meteors, since meteors are believed to have originated from the same solar nebula as Earth and offer clues about its probable internal composition.

Q 94. Ans. A

Explanation:

- **Pair 1 is correctly matched:** Alluvial soil is highly fertile and renewed frequently, making it ideal for the cultivation of **Jute** (predominantly in the flood plains of the Ganga-Brahmaputra delta in West Bengal).
- **Pair 2 is incorrectly matched:** Laterite soils are highly leached, acidic, and generally poor in organic matter. They are prominently suitable for plantation crops like **cashew nuts, tea, coffee, rubber, and cinchona**. Bananas require highly fertile, deep, well-drained loamy or alluvial soils rich in organic matter.
- **Pair 3 is incorrectly matched:** Peaty and marshy soils are heavy, dark, and highly acidic, found in areas of heavy rainfall and waterlogging. They support mangroves (like in the Sunderbans) or specific water-tolerant crops like paddy (e.g., in the Kuttanad region of Kerala). **Cashew** is predominantly grown in the Red Laterite soils of Kerala, Andhra Pradesh, and Tamil Nadu.
- **Additional Facts:**
 - ✓ **Laterite Soil Formation:** The word "laterite" comes from the Latin word later, meaning brick. These soils are formed under conditions of high temperature and heavy rainfall with alternate wet and dry spells, leading to intense leaching (washing away of siliceous matter) that leaves behind iron and aluminum oxides. When baked, this soil becomes as hard as a brick, which is why it is extensively used as a building material.
 - ✓ **Black Soil (Regur):** Found predominantly in the Deccan Trap region (Maharashtra, Gujarat, Madhya Pradesh). It is highly retentive of moisture, making it exceptionally well-suited for **cotton** cultivation. It exhibits a unique self-ploughing characteristic (developing deep cracks during dry seasons).
 - ✓ **Peaty Soils:** These occur largely in the northern part of Bihar, the southern part of Uttarakhand, and the coastal areas of West Bengal, Odisha, and Tamil Nadu. They contain large amounts of dead organic matter, sometimes making up 40-50% of the soil.

Q 95. Ans. B

Explanation:

- **Statement 1 is correct:** Extratropical cyclones generally develop along the polar front, the boundary where cold polar air meets warm tropical air in the mid-latitudes. The strong temperature contrast along this front creates atmospheric instability, leading to the formation and intensification of these cyclones.
- **Statement 2 is incorrect:** Extratropical cyclones are cold-core systems, not warm-core systems. They derive their energy from horizontal temperature differences (baroclinic instability) between contrasting air masses. In contrast, tropical cyclones are warm-core systems powered by the release of latent heat from condensation.
- **Statement 3 is incorrect:** Extratropical cyclones generally move from west to east under the influence of the prevailing westerlies in the mid-latitudes. Movement from east to west is characteristic of tropical cyclones during their initial stages due to the influence of the trade winds.

Q 96. Ans. B

Explanation:

- The Krishna River is one of the major east-flowing rivers of Peninsular India. It originates near Mahabaleshwar in the Western Ghats at an elevation of about 1,337 metres above sea level in Satara district of Maharashtra. From there, it flows eastward through Maharashtra, Karnataka, Telangana, and Andhra Pradesh before emptying into the Bay of Bengal.

Additional facts:

River	Source
Krishna	Mahabaleshwar, Western Ghats (Maharashtra)
Kaveri	Brahmagiri Hills, Western Ghats (Karnataka)
Godavari	Trimbakeshwar, Western Ghats (Maharashtra)
Narmada	Amarkantak Plateau (Madhya Pradesh)
Tapti (Tapi)	Satpura Range, Madhya Pradesh

Q 97. Ans. C

Explanation:

- **Statement 1 is correct:** Secondary (S) waves are transverse (shear) waves, in which particles of the medium vibrate perpendicular to the direction of

wave propagation. They travel slower than Primary (P) waves and are the second fastest type of seismic body waves.

- **Statement 2 is incorrect:** S-waves cannot propagate through liquids or gases because these media do not possess the shear strength required to transmit transverse waves. They can travel only through solids. This property provides evidence that the Earth's outer core is liquid, as S-waves do not pass through it.
- **Statement 3 is correct:** S-waves are recorded after P-waves on a seismograph because they travel at a lower velocity than P-waves. The difference in arrival times of P- and S-waves helps seismologists determine the location and distance of an earthquake's epicentre.

Q 98. Ans. A

Explanation:

- India is located entirely in the Northern and Eastern Hemispheres. Its mainland extends from $8^{\circ}4' \text{ N}$ to $37^{\circ}6' \text{ N}$ latitude and $68^{\circ}7' \text{ E}$ to $97^{\circ}25' \text{ E}$ longitude. The Tropic of Cancer ($23^{\circ}30' \text{ N}$) passes through the middle of the country, dividing it into almost two equal parts. Due to its latitudinal extent, India experiences a wide variety of climatic conditions, while its longitudinal extent accounts for a time difference of nearly 2 hours between its easternmost and westernmost points. However, to maintain uniformity, India follows a single Indian Standard Time (IST) based on $82^{\circ}30' \text{ E}$ longitude.

Q 99. Ans. B

Explanation:

- **Local Time:** This is the time reckoned by the apparent position of the Sun in the sky at a specific longitude. For example, local noon occurs exactly when the Sun is at its highest point in the sky above that specific meridian. Because the Earth rotates, every single degree of longitude has a different local time (a difference of 4 minutes per degree).
- **Standard Time:** Because it would be administratively chaotic for every town to operate on its own solar local time, countries adopt a "Standard Time." This is a uniform time applied across a broad geographical region or an entire country, usually based on the local time of a specific, central meridian passing through it.
- **Additional Facts:**
 - ✓ **Indian Standard Time (IST):** India has a vast longitudinal extent (roughly 68°E to 97°E),

which means the local solar time in Gujarat is nearly two hours behind the local solar time in Arunachal Pradesh. To maintain uniformity, India adopted the local time of the 82.5° E **longitude** (which passes through Mirzapur, Uttar Pradesh) as its standard time for the entire nation. IST is 5 hours and 30 minutes ahead of Greenwich Mean Time (GMT).

- ✓ **Multiple Time Zones:** Countries with massive east-to-west longitudinal extents, such as Russia, the United States, Canada, and Australia, cannot rely on a single standard time. Russia currently spans 11 different time zones.
- ✓ **Daylight Saving Time (DST):** Many countries in temperate latitudes advance their standard clocks by one hour during the summer months to make better use of extended daylight in the evenings. India does not observe DST because its equatorial proximity means the variation in daylight hours between summer and winter is relatively small.
- ✓ **International Date Line (IDL):** Located roughly at 180° longitude, this imaginary line determines where a calendar day begins and ends. Crossing it eastward subtracts a day, while crossing it westward adds a day.

Q 100. Ans. D

Explanation:

- The Grand Canyon, one of the most famous natural landmarks in the world, was carved over millions of years by the **Colorado River** together with tectonic uplift and other geomorphic processes. The river's powerful erosive force, combined with the geological uplift of the Colorado Plateau, cut deeply into the rock, creating the massive canyon we see today.
- **Additional Facts:**
 - ✓ **Geological Time Capsule:** The Grand Canyon, located in Arizona, USA, is over 277 miles long, up to 18 miles wide, and over a mile deep. The exposed rock layers in the canyon walls act as a massive geological record, revealing nearly two billion years of Earth's history.
 - ✓ **The Rio Grande:** This major river flows from Colorado to the Gulf of Mexico and is most famous for forming a significant portion of the international border between the United States (Texas) and Mexico. It carved the Rio Grande Gorge in New Mexico.

- ✓ **The Columbia River:** Flowing through the Pacific Northwest region of North America, it is the largest river in the region and is heavily dammed for hydroelectric power. It carved the spectacular Columbia River Gorge.
- ✓ **The Yukon River:** This is a major watercourse of northwestern North America, flowing through Canada's Yukon territory and the U.S. state of Alaska before emptying into the Bering Sea. It was famously used as the principal route during the Klondike Gold Rush.

Q 101. Ans. B

Explanation:

- Under Article 231, Parliament may establish a common High Court for two or more states, or for two or more states and a union territory, by law; examples include the common High Court for Punjab and Haryana at Chandigarh and the common High Court for the north eastern states at Guwahati.
- Article 214 provides that there shall be a High Court for each state, but Article 231 creates the exception permitting a common High Court to serve multiple states. Article 217 governs the appointment of High Court judges by the President, and Article 226 confers the wide writ jurisdiction of a High Court for the enforcement of fundamental rights and for any other purpose.

Q 102. Ans. C

Explanation:

- **Assertion is true:** Permanent Lok Adalats, established for public utility services under a 2002 amendment to the Legal Services Authorities Act, possess adjudicatory powers and can decide a dispute on its merits even where the parties fail to reach a settlement, up to a prescribed pecuniary limit.
- **Reason is false:** An award of a Permanent Lok Adalat is in fact final and binding on the parties and on persons claiming under them, and it cannot be appealed before a civil court, just as an ordinary Lok Adalat award is deemed to be a decree of a civil court and admits of no appeal.
- This adjudicatory power distinguishes Permanent Lok Adalats from ordinary Lok Adalats, which can only facilitate a compromise and cannot impose a decision when the parties fail to agree. Both mechanisms operate under the framework of the National Legal Services Authority set up under

the 1987 Act, alongside related institutions such as Family Courts under the 1984 Act and Gram Nyayalayas under the 2008 Act.

Q 103. Ans. C

Explanation:

- **Pair 1 is correctly matched:** The doctrine of pleasure under Article 310 means that a civil servant holds office during the pleasure of the President or the Governor, subject to the safeguards in Article 311.
- **Pair 2 is incorrectly matched:** The doctrine of basic structure applies to constitutional amendments made under Article 368, not to ordinary laws made by Parliament, which are tested instead against specific constitutional provisions such as the fundamental rights and legislative competence.
- **Pair 3 is correctly matched:** The doctrine of severability, read into Article 13, means that only the part of a law that is inconsistent with the fundamental rights becomes void, while the rest of the law, if separable, continues to remain valid.
- **Pair 4 is correctly matched:** The doctrine of eclipse holds that a pre constitution law inconsistent with fundamental rights is not wiped out but remains eclipsed, becoming dormant rather than dead, and can revive if the relevant fundamental right is subsequently amended, so three of the four pairs are correctly matched.

Q 104. Ans. B

Explanation:

- **Pair 1 is correct:** Citizenship by naturalisation is granted to a qualified foreigner, who is not an illegal migrant, on application, after fulfilling residence and other conditions and taking an oath of allegiance to the Constitution.
- **Pair 2 is correct:** Citizenship by registration is indeed one of the five modes recognised under the Citizenship Act of 1955, alongside citizenship by birth, descent, naturalisation, and incorporation of territory.
- **Pair 3 is correct:** Termination of citizenship occurs automatically when a citizen voluntarily acquires the citizenship of another country, consistent with India's principle of single citizenship, which does not permit dual citizenship.
- **Pair 4 is incorrect:** An Overseas Citizen of India is not entitled to vote in elections to the Lok Sabha or any other legislature, since such a person lacks political rights and is not a full citizen of India, unlike a Non-Resident Indian who, being a citizen, retains the right to vote.

Q 105. Ans. C

Explanation:

- The writ of quo warranto, meaning by what authority, is issued to inquire into the legality of a person's claim to a substantive public office, and unlike the other writs it can be sought by any interested person and not necessarily by the party whose own right has been infringed.
- Mandamus commands a public official or body to perform a public duty it has failed to perform, certiorari is issued by a higher court to quash an order already passed by a lower court or tribunal, and prohibition is issued to prevent a lower court or tribunal from exceeding its jurisdiction while proceedings are still pending, thereby operating at an earlier stage than certiorari.

Q 106. Ans. B

Explanation:

- **Assertion is true:** NITI Aayog does not have the power to allocate plan funds to the states, a function that its predecessor, the Planning Commission, used to perform; NITI Aayog instead functions primarily as a policy think tank offering advice and knowledge support to the Centre and the states.
- **Reason is true but does not correctly explain the Assertion:** NITI Aayog is indeed a non constitutional and non statutory body, established by an executive resolution in 2015, but this status by itself does not explain why it lacks the power to allocate funds; the absence of fund allocation power is a deliberate design choice to shift its role away from that of the Planning Commission towards advisory functions, and other non statutory bodies can and do control finances, so the extra constitutional nature is not the correct explanation.
- NITI Aayog, or the National Institution for Transforming India, is chaired by the Prime Minister and includes a Governing Council comprising the chief ministers of all states and union territories with a legislature. It replaced the Planning Commission with the aim of fostering cooperative federalism through structured support and mutual consultation rather than top down fund allocation.

Q 107. Ans. B

Explanation:

- The Central Bureau of Investigation is not established by a dedicated CBI Act; it functions

under a Home Ministry resolution of 1963, but its actual legal power to investigate offences is derived from the Delhi Special Police Establishment Act of 1946, a pre constitution law originally meant to investigate bribery and corruption among war supply contracts.

- Because its investigative jurisdiction rests on the Delhi Special Police Establishment Act, the CBI requires the consent of the concerned state government to conduct an investigation within that state, except when directed by the Supreme Court or a High Court. The Prevention of Corruption Act of 1988 is a substantive law the CBI often invokes while investigating corruption cases, and the CVC Act of 2003 relates to the Central Vigilance Commission's superintendence over the CBI in corruption matters, not to the CBI's own founding power.

Q 108. Ans. B

Explanation:

- **Pair 1 is correctly matched:** A hung assembly refers to a situation where no single party or pre poll alliance secures a clear majority of seats in a legislature, often necessitating post poll coalitions or support arrangements to form a government.
- **Pair 2 is incorrectly matched:** The anti-defection law under the Tenth Schedule applies to members belonging to a political party and to nominated and independent members in specified circumstances; it is not confined only to independent members, since its central concern is defection by members elected on a party ticket.
- **Pair 3 is correctly matched:** Coalition dharma is a term popularly used to describe the compromises, accommodations, and give and take that coalition partners must observe to sustain a coalition government.
- **Pair 4 is incorrectly matched:** A pressure group, by definition, does not directly contest elections under its own banner; it seeks to influence policy from outside the electoral process, which distinguishes it from a political party, so only two of the four pairs are correctly matched.

Q 109. Ans. A

Explanation:

- **Statement 1 is correct:** The Competition Commission of India (CCI) is a statutory body of the Government of India. It was established under the **Competition Act, 2002**, and was duly constituted in March 2009.

- **Statement 2 is incorrect:** Appeals against the orders or decisions of the CCI are not filed directly before the Supreme Court. They are first heard by the **National Company Law Appellate Tribunal (NCLAT)**. (Prior to 2017, appeals went to the Competition Appellate Tribunal or COMPAT, but its functions were subsequently merged into NCLAT). If a party is aggrieved by the NCLAT's decision, they can then appeal to the Supreme Court.

- **Additional Facts:**

- ✓ **Composition:** The CCI consists of a Chairperson and a maximum of six Members, who are appointed by the Central Government.
- ✓ **Primary Mandate:** Its main objectives are to eliminate practices having an adverse effect on competition, promote and sustain competition, protect the interests of consumers, and ensure freedom of trade in the markets of India.
- ✓ **Historic Shift:** The Competition Act of 2002 replaced the archaic **Monopolies and Restrictive Trade Practices (MRTP) Act of 1969**. The shift was driven by the economic liberalization of 1991, moving away from curbing monopolies to actively promoting healthy market competition.
- ✓ **Suo Moto Powers:** The CCI has the power to initiate an inquiry suo moto (on its own motion) if it suspects any enterprise of abusing its dominant position or entering into anti-competitive agreements.

Q 110. Ans. A

Explanation:

- **Statement 1 is correct:** According to Article 324(5) of the Constitution, the Chief Election Commissioner (CEC) can only be removed from office in the same manner and on the same grounds as a Judge of the Supreme Court. Article 124(4) specifies that a Supreme Court Judge can only be removed on the grounds of "**proved misbehaviour or incapacity**."
- **Statement 2 is correct:** Because the removal process mirrors that of a Supreme Court Judge, it is governed by the Judges (Inquiry) Act, 1968. Under this Act, a removal motion must be supported by the signatures of at least **100 members** if introduced in the Lok Sabha (or 50 members if introduced in the Rajya Sabha).
- **Statement 3 is incorrect:** The Lok Sabha Speaker (or the Rajya Sabha Chairman, depending on where

the motion is introduced) is **not bound** to admit the notice. The presiding officer has the absolute discretionary power to either admit the motion or refuse to admit it, even if it has the requisite number of signatures.

- **Additional Facts:**

- ✓ **Protection of Other Commissioners:** While the CEC enjoys security of tenure akin to a Supreme Court Judge, the other Election Commissioners (ECs) do not have the exact same constitutional protection against removal. They can be removed by the President, but only upon the recommendation of the Chief Election Commissioner.
- ✓ **Special Majority Requirement:** If the Speaker/Chairman admits the removal motion, a three-member inquiry committee is formed. If the committee finds the CEC guilty, Parliament must pass the motion with a **special majority** in both houses (a majority of the total membership of the House and a majority of not less than two-thirds of the members present and voting) in the same session before the President issues the removal order.
- ✓ **Article 324:** This is the bedrock article that vests the superintendence, direction, and control of all elections to Parliament, State Legislatures, and the offices of the President and Vice-President in the Election Commission.

Q 111. Ans. C

Explanation:

- **Statement 1 is correct:** Qatar is the second-largest producer of helium in the world (after the United States) and accounts for roughly one-third (approx. 30–35%) of the global helium supply. It extracts helium as a byproduct of its liquefied natural gas (LNG) processing.
- **Statement 2 is correct:** India relies entirely (100%) on imports to meet its commercial helium requirements, with the vast majority imported from Qatar and the US. While Indian researchers have traced minor helium reserves in geothermal springs (such as Bakreshwar in West Bengal), they are not currently exploited for commercial production.
- **Statement 3 is incorrect:** Helium is a **strictly non-renewable** resource. On Earth, it is generated incredibly slowly over billions of years through the natural radioactive decay of heavy elements (like uranium and thorium) deep within the crust. Because it is so light and unreactive, once released into the

atmosphere, it escapes Earth's gravity and is lost to space forever. Furthermore, it cannot be chemically synthesised in laboratories (creating it requires nuclear fusion, like the processes inside the Sun).

- **Statement 4 is correct:** Liquid helium has an incredibly low boiling point (-269°C or 4.2 Kelvin). This makes it the only element capable of safely cooling the heavy superconducting magnets inside medical MRI (Magnetic Resonance Imaging) machines to the extreme temperatures required for them to function without electrical resistance. The medical sector is the largest single consumer of global helium.

- **Additional Facts:**

- ✓ **Qatar (Ras Laffan Industrial City):** Ras Laffan is the world's largest LNG plant and helium production base. In March 2026, the facility suffered extensive damage from Iranian drone strikes during the US-Iran conflict. This single attack forced QatarGas to declare force majeure, instantly halting $\sim 30\%$ of the global helium supply and causing severe disruptions in semiconductor manufacturing.
- ✓ **Russia (Orenburg & Amur):** The **Orenburg Helium Plant** (Russia's primary established helium extraction facility) and its adjacent gas processing plant were struck by long-range Ukrainian drones in June 2026. Additionally, Russia operates the massive **Amur Gas Processing Plant** in its Far East, which is critical to its export market.
- ✓ **United States (Amarillo, Texas):** Home to the Federal Helium Reserve and the Cliffside Gas Field, historically the most important helium storage and processing infrastructure in the world.
- ✓ **Algeria (Skikda and Arzew):** Major African industrial port cities that serve as primary LNG and helium processing hubs exporting to Europe.

Q 112. Ans. B

Explanation:

- The "Yellow Line" is an informal, self-declared military boundary created by Israel. It serves as a unilateral, militarised buffer zone inside conflict territories to establish a controlled security perimeter.
- The strategy first emerged during the 2025 Gaza war to separate Israeli-controlled areas from Palestinian-held territories. In 2026, the Israel Defence Forces (IDF) extended this strategy into southern Lebanon, pushing up to the Litani River during a temporary ceasefire.

- Unlike legally recognised international borders (like the UN-demarcated "Blue Line"), the Yellow Line is a fluid, forward defensive posture. It is designed to prevent militant groups such as Hamas and Hezbollah from re-establishing operational strength near Israeli population centres.

Q 113. Ans. C

Explanation:

- **The Union Budget 2026–27 announced the establishment of Dedicated Rare Earth Corridors exclusively in four states: Odisha, Kerala, Andhra Pradesh, and Tamil Nadu.** Therefore, Karnataka is not a part of this specific initiative.
- These corridors are designed to integrate the entire value chain of critical minerals, encompassing mining, processing, research, and the domestic manufacturing of Rare Earth Permanent Magnets (REPMs).
- **Additional Facts:**
 - ✓ **The selected states:** Odisha, Kerala, Andhra Pradesh, and Tamil Nadu, share extensive coastlines that contain India's richest **placer beach sand deposits**. These sands are abundant in **Monazite**, the primary radioactive ore used to extract Light Rare Earth Elements (LREEs) in India.
 - ✓ The primary industrial focus of these corridors is the manufacturing of **Rare Earth Permanent Magnets (REPMs)**. The most critical of these are Neodymium-Iron-Boron (NdFeB) and Samarium-Cobalt (SmCo) magnets.
 - ✓ REPMs are essential components for modern energy and defence transitions. They are heavily utilised in manufacturing electric vehicle (EV) motors, direct-drive wind turbine generators, aerospace guidance systems, and advanced consumer electronics.
 - ✓ In India, the processing of Monazite is strictly controlled due to its thorium content. **Indian Rare Earths Limited (IREL)**, a Public Sector Undertaking under the Department of Atomic Energy (DAE), is the sole nodal agency permitted to mine and process it.

Q 114. Ans. B

Explanation:

- **Statement 1 is correct:** The ASISSE is the first-ever survey of its kind launched by the National Statistics Office (NSO), under the Ministry of Statistics and

Programme Implementation (MoSPI). It uses the GSTN (Goods and Services Tax Network) database as its sampling frame to identify the enterprises.

- **Statement 2 is incorrect:** ASISSE exclusively covers incorporated corporate entities in the services sector (specifically, companies registered under the Companies Act of 1956/2013 or the Limited Liability Partnership Act, 2008). The unincorporated non-agricultural sector is covered separately by a different survey called the Annual Survey of Unincorporated Sector Enterprises (ASUSE).
- **Statement 3 is correct:** The survey operates with legal backing under the Collection of Statistics Act, 2008 (amended in 2017 and by the Jan Vishwas Act, 2023). The reference period for this first-ever survey is the financial year 2024–25.
- **Additional Facts:**
 - ✓ The ASISSE will survey approximately 1.21 lakh enterprises across all States and Union Territories. Data will be collected strictly through a secure web-based portal.
 - ✓ Alongside the launch, the NSO released a reader-friendly explainer document titled “Know Your Survey: A User Guide to the ASISSE.” It is designed to promote transparency, explain reporting requirements, and increase the ease of compliance for participating enterprises.
 - ✓ **MoSPI now utilises three major surveys to capture the complete non-agricultural economy:**
 - ASI (Annual Survey of Industries) → Covers the registered manufacturing sector.
 - ASUSE (Annual Survey of Unincorporated Sector Enterprises) → Covers unincorporated non-agricultural entities.
 - ASISSE → Covers the incorporated services sector (trade, IT, health, hospitality, etc.).

Q 115. Ans. C

Explanation:

- According to NITI Aayog's Fiscal Health Index 2026, **Jharkhand** successfully secured a position in the top “Achievers” category among the 18 major states, alongside Odisha and Goa.
- **Punjab & West Bengal:** These states are placed in the “Aspirational” (bottom performers) category. They continue to face elevated debt, high interest commitments, and struggle with persistent fiscal stress.

- **Bihar:** Bihar improved its position from the Aspirational category to the “Performer” category, signaling better deficit management, but it has not reached the top Achievers category.

Additional Facts:

- ✓ **Five Pillars:** The index evaluates the financial health of Indian states across five key pillars: Quality of Expenditure, Revenue Mobilisation, Fiscal Prudence, Debt Index, and Debt Sustainability.
- ✓ **Top Performer:** **Odisha** continues to lead the rankings as the top Achiever for the second consecutive year, driven by controlled deficits, high own-tax shares, and stable revenues.
- ✓ **Special Category Expansion:** For the first time, the 2026 edition separately evaluated 10 North-Eastern and Himalayan states. In this group, **Arunachal Pradesh** and **Uttarakhand** emerged as the top “Achievers.”
- ✓ **Data Source:** The index uses data verified by the Comptroller and Auditor General (CAG) to ensure rigor, credibility, and transparency in assessing the fiscal federalism of the states.

Q 116. Ans. C

Explanation:

- **Nafithromycin** is the first indigenously developed antibiotic by India. It was formally launched in November 2024 to combat the growing global threat of Antimicrobial Resistance (AMR). Developed by the pharmaceutical company Wockhardt with the support of the Biotechnology Industry Research Assistance Council (BIRAC), it is marketed under the brand name “Miqnaf.”
- **Penicillin:** Discovered by Scottish researcher Alexander Fleming in 1928, this was the world's first widely used antibiotic.
- **Rifamycin:** Discovered in 1957 by a research team at Lepetit Pharmaceuticals in Italy.
- **Tetracycline:** Discovered by American botanist Benjamin Minge Duggar in 1945.
- **Additional Facts:**
 - ✓ **Target Disease:** Nafithromycin is a next-generation macrolide antibiotic specifically designed to treat Community-Acquired Bacterial Pneumonia (CABP) caused by drug-resistant bacteria. It is especially useful for vulnerable populations, including the elderly, children, and patients with compromised immune systems (such as those with diabetes or cancer).

- ✓ **Global Breakthrough:** It marks a historic milestone as the first new antibiotic in its class to be successfully developed and introduced anywhere in the world in over 30 years.
- ✓ **High Efficacy:** Clinical trials have shown that Nafithromycin is ten times more effective against certain pathogens than current standard treatments like azithromycin. It achieves strong outcomes with just a short, three-day treatment regimen, which helps reduce the chance of patients abandoning their medication early (a major driver of antibiotic resistance).

Q 117. Ans. A

Explanation:

- **Bank of Baroda (BoB)** recently became the first bank in India to launch a UPI-linked overdraft (OD) facility specifically for women Self-Help Group (SHG) members.
- This digital micro-credit innovation provides eligible women SHG members holding Pradhan Mantri Jan Dhan Yojana (PMJDY) accounts with instant access to an overdraft limit of up to **₹5,000**.
- **Additional Facts:**
 - ✓ **Branchless Access:** The initiative completely eliminates the need for women to visit a bank branch. They can link their SHG OD account to any UPI-enabled app (like Google Pay, PhonePe, or BHIM) to access credit, make merchant QR payments, and repay their dues digitally.
 - ✓ **Key Partners:** The facility was developed and launched in collaboration with the National Payments Corporation of India (NPCI), the Indian Banks' Association (IBA), and the Deendayal Antyodaya Yojana-National Rural Livelihoods Mission (DAY-NRLM).
 - ✓ **Financial Inclusion:** By embedding overdraft facilities directly into UPI, this initiative helps rural women build a formal credit footprint, making them eligible for higher-value loans in the future while strengthening women-led grassroots economic growth.

Q 118. Ans. B

Explanation:

- India's first green methanol production facility is being established at the Deendayal Port Authority (DPA) in Kandla, Gujarat. The plant utilises invasive biomass, specifically the *Prosopis juliflora* shrub, as feedstock to produce green methanol for use as a clean marine fuel.

Q 119. Ans. D

Explanation:

- **Aoleang Monu (Nagaland):** This is the premier spring festival of the Konyak Naga tribe, celebrated in the first week of April. It marks the end of the sowing season, welcomes the new year, and involves vibrant traditional dances and feasts.
- **Suvori (Suwari) Festival (Assam):** An indigenous spring festival celebrated in Assam (particularly in the Boko region) that coincides with Rongali Bihu. It is renowned for its mass Bihu performances and traditional sports like elephant and horse races, bringing together diverse communities such as the Rabha, Boro, and Garo.
- **Gangamma Jatara (Andhra Pradesh):** A major annual folk festival held predominantly in Tirupati. It honors the village deity, Sri Tathaya Gunta Gangamma, and is famous for its unique rituals where devotees, particularly men, don elaborate costumes and face paint (veshams).
- **Etor Gidi (Arunachal Pradesh):** A traditional agro-based festival celebrated by the Adi community in mid-May. It marks the beginning of the farming season and involves the communal repairing of fences around agricultural fields to protect crops from livestock, alongside the performance of the traditional Delong dance.
- **Additional Facts:**
 - ✓ **The Adi Tribe:** One of the most populous indigenous groups in Arunachal Pradesh, they are known for their intricate weaving, cane and bamboo crafts, and traditional village councils called Kebang, which handle local dispute resolution.
 - ✓ **The Konyak Nagas:** Recognized as the largest of the Naga tribes, they inhabit the Mon district of Nagaland. They are historically famous for their fierce headhunting traditions and distinct, elaborate facial and body tattoos.
 - ✓ **Dere or Musup:** In the Adi community, the Dere (or Musup) serves as the central community hall or youth dormitory where vital social, political, and cultural activities—like the Etor Gidi gatherings—take place.

Q 120. Ans. A

Explanation:

- **Statement 1 is correct:** The 38th National Games were successfully hosted by **Uttarakhand**. The event officially commenced on January 28, 2025,

and concluded on February 14, 2025, featuring over 11,000 athletes from across India competing in 38 sports disciplines spread across multiple cities in the state.

- **Statement 2 is correct:** The official theme of the event was “Green Games,” emphasizing environmental sustainability and reflecting the pristine natural beauty of Uttarakhand. Additionally, the official torch of the Games was named **Tejaswini** (meaning radiant and strong), representing power, inspiration, and the relentless pursuit of excellence. (The official tagline for the games was “Sankalp Se Shikhar Tak”).
- **Statement 3 is incorrect:** The mascot of the 38th National Games was not Nanda Devi. The mascot was named **Mauli**, which was inspired by the **Himalayan Monal**, the vibrant state bird of Uttarakhand. The mascot was chosen to symbolize the region’s rich biodiversity, natural beauty, and the enduring spirit of its people.
- **Additional Facts:**
 - ✓ The National Games, originally known as the Indian Olympic Games, first took place in Lahore in 1924. They are an Olympic-style multi-sport event organized by the Indian Olympic Association (IOA) where athletes from various states and Union Territories compete for medals.
 - ✓ The 38th edition notably included traditional and indigenous sports as demonstration events, such as Kalaripayattu, Mallakhamb, and Yogasana, to promote India’s cultural sporting roots.
 - ✓ The 37th National Games were held in Goa in 2023, and the 39th National Games are scheduled to be hosted by Meghalaya.

Q 121. Ans. A

Explanation:

- **Pair 1 is correctly matched:** Dr. Ch. Srinivasa Rao (Director and Vice-Chancellor of ICAR-Indian Agricultural Research Institute) was conferred with the 9th Prof. M.S. Swaminathan Award in April 2026 for his outstanding contributions to climate-resilient agriculture and sustainable natural resource management in India.
- **Pair 2 is correctly matched:** Indian actress and producer Sayani Gupta was named the Harvard South Asian “Person of the Year” 2026 by the Harvard University South Asian Association. She was honored for her meaningful contributions to

authentic storytelling and her role in shaping South Asian representation on global platforms.

- **Pair 3 is incorrectly matched:** The Machchhli Award (often instituted by wildlife NGOs or state authorities like the Rajasthan government) is a conservation award named after the legendary Ranthambore tigress, Machchhli (T-16), who was once the world’s most photographed tiger. It was awarded to Anita Chaudhary, a forest guard posted at Rajasthan’s Shergarh sanctuary.

Q 122. Ans. A

Explanation:

- **Mukhi** became the first India-born cheetah to successfully reproduce, giving birth to five healthy cubs at Kuno National Park in Madhya Pradesh.
- **Project Cheetah Milestone:** Mukhi was born in March 2023 to a Namibian cheetah (Jwala) as part of India’s ambitious Cheetah Reintroduction Project. Her transition from a hand-reared cub (she was the sole survivor of her litter after her siblings succumbed to extreme summer heat) to a mother is being hailed as a historic milestone, proving that the big cats can successfully adapt, thrive, and breed across generations in Indian habitats.
- **Genetic Diversity:** The birth of five cubs is an unusually large litter for cheetahs and offers significant hope for establishing a self-sustaining and genetically diverse cheetah population in India, seven decades after they were declared extinct in the country.
- **Additional Facts:**
 - ✓ **Translocation:** Project Cheetah marks the world’s first inter-continental large wild carnivore translocation project. The first batch of cheetahs was brought from Namibia to Kuno National Park in September 2022.
 - ✓ **Kuno National Park:** Located in the Vindhyan Hills of Madhya Pradesh, Kuno was selected for the reintroduction due to its suitable grassland and savannah habitat, which closely mirrors the cheetah’s natural African environment, and its adequate prey base (such as chital).

Q 123. Ans. C

Explanation:

- **Statement 1 is correct:** The first phase of the All India Tiger Estimation 2026 (AITE-26), a massive exercise coordinated by the National Tiger Conservation Authority (NTCA) and the Wildlife Institute of India (WII), indeed commenced at

the Thanthai Periyar Wildlife Sanctuary in Tamil Nadu. This site is significant as it forms a critical wildlife corridor connecting the Sathyamangalam Tiger Reserve and the Male Mahadeshwara Hills Tiger Reserve.

- **Statement 2 is correct:** Kerala declared *Bacillus subtilis* as its State Microbe in January 2026. This historic decision makes Kerala the first Indian state to formally recognise a microorganism in its roster of official state symbols. The announcement was made alongside the launch of the Centre of Excellence in Microbiome (CoEM) to highlight the bacterium's beneficial role in sustainable agriculture, biotechnology, and soil health.

Q 124. Ans. A

Explanation:

- **500 GW Renewable Energy (2030):** India committed to achieving 500 GW of non-fossil fuel energy capacity by 2030 as part of its climate action targets announced at COP26.
- **India's Space Station (2035):** The Indian Space Research Organisation (ISRO) has set a firm timeline to construct and commission the preliminary modules of the Bharatiya Antariksha Station by 2035.
- **Manned Moon Mission (2040):** Following the success of the Chandrayaan missions, Prime Minister Modi directed ISRO to aim for landing an Indian astronaut on the Moon by 2040.
- **Developed Nation / Viksit Bharat (2047):** This vision targets Transforming India into a completely developed economy by 2047, coinciding with the centenary celebration of India's independence.

Q 125. Ans. B

Explanation:

- The 4th India–Central Asia Dialogue, held in New Delhi on June 6, 2025, was chaired and hosted by India's External Affairs Minister, **Dr. S. Jaishankar**.
- The dialogue brought together the Foreign Ministers of the five Central Asian countries, Kazakhstan, the Kyrgyz Republic, Tajikistan, Turkmenistan, and Uzbekistan to discuss regional security, economic cooperation, and connectivity. A key focus of the meeting was enhancing digital public infrastructure (DPI), with India agreeing to provide assistance in developing these systems in Central Asian countries. The meeting also addressed shared concerns regarding regional stability, counter-terrorism, and the humanitarian situation in Afghanistan.

Q 126. Ans. B

Explanation:

- **Pair 1 is correctly matched:** Magadi Kere Conservation Reserve is a 54.4-hectare human-made wetland located in **Shirahatti Taluk, Gadag district, Karnataka**. It was declared a Conservation Reserve under the Wildlife (Protection) Act, 1972, on February 14, 2023, and simultaneously designated a **Ramsar site** (Site No. 2536). It is an Important Bird and Biodiversity Area (IBA) and one of the largest wintering grounds for the bar-headed goose in South India.
- **Pair 2 is incorrectly matched:** Longwood Shola Reserve Forest is located in **Kotagiri, Nilgiris district, Tamil Nadu**, not Kerala. It spans 116 hectares in the Nilgiri Hills at the junction of the Eastern and Western Ghats. It was designated a Ramsar site in 2024 and is considered one of the richest sites for amphibian diversity in Asia.
- **Pair 3 is correctly matched:** Shallabugh (Shalbug) Wetland Conservation Reserve is located in **Ganderbal district, Jammu and Kashmir**, spread across 1,675 hectares in the Sindh river delta, west of Anchar Lake. It was declared a Ramsar site on **June 8, 2022**, along with Hygam Wetland. It is a critical habitat for over 21 species of resident and migratory birds including the endangered Steppe Eagle and Pallas's Fish-Eagle.

Q 127. Ans. C

Explanation:

- **The IRF Summit Africa 2025, titled "Africa Together: A Continental Call to Religious Freedom"**, was hosted at the **Safari Park Hotel, Nairobi**, and was a joint initiative of the IRF Summit, **Pepperdine University**, the **Religious Freedom Institute (RFI)**, and the **Global Peace Foundation (GPF)**.
- It was the **first regional edition** of the IRF Summit to be held on the **African continent**, bringing together Chief Justices, UN representatives, faith leaders, human rights lawyers, and civil society groups from across Africa.
- The focus was on the growing threat of **religiously-motivated terrorism** in Africa and government restrictions on freedom of religion or belief (FoRB), particularly in Sub-Saharan Africa.
- The IRF Summit was established under **Sam Brownback**, former US Ambassador-at-Large for International Religious Freedom, and is co-chaired with **Katrina Lantos Swett**.

Q 128. Ans. A

Explanation:

- The Bihar Cabinet, at a meeting chaired by Chief Minister **Samrat Choudhary** on **May 20, 2026**, approved a proposal allowing **IndiGo Airlines (InterGlobe Aviation Limited)** to operate non-stop international flights on the **Gaya–Bangkok** route for a period of **12 months**.
- IndiGo was selected on a **nomination basis** as it was the sole bidder for the route. The state government will provide **Rs 10.40 crore as Viability Gap Funding (VGF)** to support the airline during the initial phase.
- The primary rationale is promotion of **Buddhist tourism**. Every year, a large number of **Thai Buddhist pilgrims** visit Bodh Gaya and other Buddhist heritage sites in Bihar. Direct air connectivity is expected to significantly increase tourist footfall, boosting transport, hospitality, handicraft, and local trade.
- **Gaya International Airport** already handles international pilgrimage traffic, particularly during the **Kalachakra** and other Buddhist festivals. This route further cements Bihar's position as a key destination on the **Buddhist Circuit** under India's tourism policy.
- **For BPSC context:** Bihar has also announced an international aviation scheme providing VGF of **Rs 5 to 10 lakh per flight** for routes connecting Bihar to Kathmandu, Singapore, Colombo, and Sharjah, but Cabinet approval for Gaya–Bangkok specifically makes Option 1 the correct answer here.

Q 129. Ans. A

Explanation:

- Renowned travel writer and naval veteran **Commander Hugh Gantzer** passed away on **February 2, 2026**, at his residence '**Oakbrook**' on Kinraig Library Road, **Mussoorie**, at the age of 94/95. He was born on **January 9, 1931, in Patna, Bihar**.
- He served in the **Indian Navy**, retiring at the rank of **Commander**, and also served as Judge Advocate at the **Southern Naval Command**. After retirement, he settled permanently in Mussoorie.
- Together with his wife **Colleen Gantzer** (who predeceased him in November 2024), he authored **over 30 books**, wrote thousands of travel articles, and produced **52 travel documentaries** broadcast on **Doordarshan**, pioneering modern travel writing in India with a focus on lesser-known destinations, local traditions, and cultural heritage.

- The couple received the **Padma Shri in 2025**, six national tourism awards, the **National Tourism Lifetime Achievement Award**, and two gold awards from the **Pacific Asia Travel Association (PATA)**. Due to ill health, Gantzer received the Padma Shri at his home from the Uttarakhand Home Secretary.
- He was also a member of the **Supreme Court-appointed monitoring committee** overseeing environmental issues in the Doon Valley and Mussoorie from 1990 onward.

Q 130. Ans. D

Explanation:

- **a. Third Chief of Defence Staff → 4 (N.S. Raja Subramani):** Lieutenant General N.S. Raja Subramani (Retd.) took over as India's third CDS on 31 May 2026, succeeding General Anil Chauhan (the second CDS), following General Bipin Rawat, India's first CDS.
- **b. Navy Chief → 1 (Krishna Swaminathan):** Vice Admiral Krishna Swaminathan, previously Flag Officer Commanding-in-Chief of the Western Naval Command, took over as the 27th Chief of the Naval Staff on 31 May 2026, succeeding Admiral Dinesh Kumar Tripathi.
- **c. Vice-Chairperson, NITI Aayog → 2 (Ashok Kumar Lahiri):** Economist Ashok Kumar Lahiri was appointed Vice-Chairperson of NITI Aayog, notified on 24 April 2026, succeeding Suman K. Bery, alongside new full-time members including Rajiv Gauba and Prof. K.V. Raju.
- **d. FOC-in-C, Western Naval Command → 3 (Sanjay Vatsyayan):** Vice Admiral Sanjay Vatsayan, who had been serving as the 47th Vice Chief of the Naval Staff, took charge as the new The Flag Officer Commanding-in-Chief (FOC-in-C) of the Western Naval Command on 30 May 2026, filling the post vacated by Krishna Swaminathan's elevation to Navy Chief.

Q 131. Ans. C

Explanation:

- On **April 11, 2026**, the first-ever life-size statue of **Swami Vivekananda** in the United States was unveiled at **Westlake Square, downtown Seattle**, jointly by **Seattle Mayor Katie Wilson** and **Consul General of India in Seattle, Prakash Gupta**. The unveiling was held on the occasion of **ICCR Day**.
- The life-size bronze statue was **gifted by the Indian Council for Cultural Relations (ICCR)** to the

City of Seattle, and sculpted by renowned Indian artist **Naresh Kumar Kumawat**, who has over 600 installations across 80 countries. Seattle thus became the **first city government in the US** to host a life-size statue of Swami Vivekananda.

- Westlake Square was chosen strategically for its extremely high footfall of approximately **4 lakh visitors daily** and its proximity to the **Amazon headquarters**, the **Seattle Convention Centre**, and major commercial hubs.
- The event is significant from the perspective of **India's cultural soft power and diplomacy**, reflecting the philosophy of **Vasudhaiva Kutumbakam** (The World is One Family). It also commemorates the spirit of Vivekananda's landmark address at the **World's Parliament of Religions in Chicago in 1893**, where he opened with the famous words "**Sisters and Brothers of America**", introducing Vedanta and Yoga to the Western world.

Q 132. Ans. A

Explanation:

- On **May 4, 2026**, Bengaluru-based space technology firm **Pixxel** announced a strategic partnership with AI startup **Sarvam** to develop "**Pathfinder**", India's first **Orbital Data Centre (ODC) satellite**. The satellite is slated for launch in **Q4 2026** and will weigh approximately **200 kg**.
- Under the partnership, **Pixxel** will design, build, launch, and operate the satellite, while **Sarvam** will provide the **AI backbone**, running full-stack language models directly onboard for both training and inference in orbit.
- **Pathfinder** will carry **data centre-grade GPUs** and a **hyperspectral imaging camera**. Unlike conventional satellites that transmit raw data to Earth, it will process and analyse data in orbit and transmit only actionable insights, reducing dependence on terrestrial cloud infrastructure.
- The satellite will operate in a **dawn-dusk Sun-Synchronous Orbit (SSO)**, enabling continuous, uninterrupted solar power generation. Key applications include crop disease detection, environmental monitoring, mineral exploration, and disaster response.
- The mission advances India's **technological sovereignty** by ensuring that India-built AI models run on India-built satellites without dependence on **foreign cloud infrastructure**, reflecting the principle of **Sovereign AI**.

Q 133. Ans. D

Explanation:

- **One Health Summit (France)**: Held on **April 5-7, 2026, in Lyon, France**, co-chaired by French President Emmanuel Macron and Ghana's President John Mahama under France's G7 Presidency. It was the ninth edition of the One Planet Summits, bringing together Heads of State for the first time to address interconnected human, animal, and environmental health threats under the One Health approach.
- **9th Indian Ocean Conference (Mauritius)**: Held on **April 10-12, 2026, in Port Louis, Mauritius**, organised by the India Foundation in association with the Government of Mauritius. It was the first edition of the IOC held on African soil. The theme was "Collective Stewardship for Indian Ocean Governance." India's EAM S. Jaishankar represented India.
- **SCO Defence Ministers' Meeting (Kyrgyzstan)**: Held on **April 28, 2026, in Bishkek, Kyrgyzstan**, under the Kyrgyz Republic's chairmanship (2025-26), with the theme "25 Years of the SCO: Together Towards Sustainable Peace, Development, and Prosperity." India was represented by Raksha Mantri **Rajnath Singh**.
- **79th World Health Assembly (Switzerland)**: Held on **May 18-23, 2026, in Geneva, Switzerland**, at the Palais des Nations. The World Health Assembly is the supreme decision-making body of WHO, held annually in Geneva. It sets global health policy and oversees WHO's budget and leadership.

Q 134. Ans. C

Explanation:

- **Pair 1 is correctly matched**: China overtook the United States to become India's largest trading partner in FY26, with bilateral trade reaching \$151.1 billion, ending the US's four-year run at the top. India's trade deficit with China hit a record \$112.16 billion.
- **Pair 2 is correctly matched**: Venezuela has emerged as India's third-largest supplier of crude oil, reflecting India's diversification of energy imports and rising purchases from Latin American producers.
- **Pair 3 is correctly matched**: India has remained the world's largest recipient of remittances, with inflows around \$135.4 billion in FY25, a position it has held consistently since 2008 per World Bank data.

- **Pair 4 is incorrectly matched:** India was overtaken as the world's 5th-largest stock market by Taiwan, not Japan. As of late May 2026, Taiwan's market capitalisation climbed to \$4.95 trillion (powered by TSMC's rally), surpassing India's \$4.92 trillion — pushing India down to 6th place, behind the US, mainland China, Japan, Hong Kong, and now Taiwan.

Q 135. Ans. C

Explanation:

- **Statement 1 is correct:** India's first transgender clinic, originally known as **Mitr Clinic**, is located in **Hyderabad** (Narayanguda area). It was established in **January 2021** through a partnership between **USAID** and **Johns Hopkins University** under the **President's Emergency Plan for AIDS Relief (PEPFAR)**. It is India's first healthcare facility **entirely staffed and managed by transgender individuals**, providing gender-affirming care, HIV/AIDS treatment, mental health support, hormone therapy guidance, and legal aid. The clinic was temporarily shut down in early 2025 following the **Trump administration's freeze on USAID funding**, but subsequently reopened as **Sabrang Clinic** with renewed support from **Tata Trusts**.
- **Statement 2 is correct:** Hyderabad-based **Ananth Technologies** received approval from **IN-SPACe** (Indian National Space Promotion and Authorisation Centre) in **July 2025** to launch India's first private satellite broadband service. The company will deploy a **4-tonne geostationary (GEO) communication satellite** positioned over **35,786 km above Earth**, capable of delivering up to **100 Gbps** of broadband capacity across India. Services are expected to commence by **2028**, with an initial investment of **Rs 3,000 crore**. Unlike Starlink, OneWeb, and Amazon Kuiper that use **LEO satellites**, Ananth Technologies is taking the **GEO route**, which offers **wider territorial coverage** with a single satellite covering the entire Indian subcontinent. The company will become the first private Indian firm to offer satcom services using a **domestically-built satellite**.

Q 136. Ans. B

Explanation:

- According to the World Bank's Spring 2025 Poverty and Equity Brief, India's Gini Index stands at 25.5, placing it among the "moderately low" inequality category (Gini score between 25–30) — ranked the

4th most equal country in the world, behind only the Slovak Republic, Slovenia, and Belarus.

● Key points:

- ✓ India's Gini Index improved from 28.8 in 2011-12 to 25.5 in 2022-23, reflecting a consistent decline in consumption-based inequality over the decade.
- ✓ India's score is notably better than that of China (35.7), the United States (41.8), and every G7 and G20 nation.
- ✓ **This achievement is linked to India's poverty reduction record:** roughly 171 million people were lifted out of extreme poverty over the decade, with the extreme poverty rate (at the \$2.15/day threshold) falling from 16.2% (2011-12) to 2.3% (2022-23).

Q 137. Ans. A

Explanation:

- Bulgaria adopted the euro as its official currency on 1 January 2026, becoming the 21st member state of the eurozone, replacing its previous currency, the Bulgarian lev.
- Bulgaria had been a European Union member since 1 January 2007, and its currency (the lev) had been pegged to the euro under the Exchange Rate Mechanism II (ERM II) since 2020.
- The conversion rate was fixed at 1.95583 lev per €1, corresponding to the lev's long-standing peg rate.

Q 138. Ans. A

Explanation:

- **a. Number one in Eastern-Region rural sanitation → 3 (Bihar):** Bihar has recently topped the eastern region in a rural sanitation survey (2026), marking a significant improvement given the state's historically poor sanitation record.
- **b. First in implementing Central Government projects → 1 (Maharashtra):** Maharashtra leads India with 215 central government infrastructure projects worth approximately ₹5.5 trillion, followed by Uttar Pradesh (180 projects, ₹4.3 trillion) and Andhra Pradesh (141 projects, ₹2.1 trillion) — attributed to its strong industrial base, port connectivity, and Mumbai's status as India's financial capital.
- **c. Lowest Infant Mortality Rate → 4 (Kerala):** Kerala has consistently maintained India's lowest Infant Mortality Rate (around 6 per 1,000 live births), attributed to its strong public healthcare system and high literacy levels — a long-standing, well-established fact in Indian health indicators (SRS Bulletin data).

- d. Country topping the list of nations hit by spam calls → 2 (Indonesia): According to Truecaller's 2025 Global Spam and Fraud Report (released May 2026), Indonesia topped the list as the world's most spammed country, with 79% of unknown calls identified as spam or fraud, followed by Chile, Vietnam, Brazil, and India (5th, at 66% spam intensity).

Q 139. Ans. C

Explanation:

- **Statement 1 is correct:** India's men's football team dropped to 133rd in the FIFA Men's World Rankings released in July 2025, its lowest ranking in nine years at that point, following back-to-back losses to Thailand and Hong Kong. This slide contributed to head coach Manolo Márquez's departure from the All India Football Federation (AIFF).
- **Statement 2 is correct:** Iga Świątek won the Wimbledon Women's Singles title in 2025 – her first Wimbledon crown and first Grand Slam title on a grass court, adding to her collection of French Open titles.

Q 140. Ans. C

Explanation:

- Tesla officially entered the Indian market in July 2025, opening its first Experience Center in India at Maker Maxity, Bandra-Kurla Complex (BKC), Mumbai on July 15, 2025. With this entry, India became Tesla's 50th global market.
- Tesla launched two variants of the Model Y in India, the Standard Range Rear-Wheel Drive and the Long Range, as its initial offerings for the Indian market.
- Tesla has since progressively expanded its footprint across India, establishing Experience Centres in the following sequence: Mumbai (July 2025) → Delhi Aerocity → Gurugram (November 26, 2025, Orchid Business Park) → Bengaluru (May 14, 2026, VR Bengaluru, Whitefield) → Hyderabad (June 17, 2026, Knowledge City, HITEC City).
- Unlike the experience centres, which focus on brand exposure and customer education, the Gurugram outlet operates as a full-service hub handling sales, delivery, and servicing.

Q 141. Ans. D

Explanation:

- By using option D in the above series, we get the following pattern: R L X G J / R L X G I / R L X G H / R L X G G
- Hence, option D is correct.

Q 142. Ans. C

Explanation:

- 'all is well' is coded as 'gh, tk, ib' ----(1)
 - **Statement I:**
 - ✓ 'all tries it' is coded as 'tk, hu, mv' ----(2)
 - ✓ 'tap is good' is coded as 'gh, qw, xz' ----(3)
 - From (1) and (2)
 - ✓ 'all' is coded as 'tk'
 - From (1) and (3)
 - ✓ 'is' is coded as 'gh'
 - Hence 'well' is coded as 'ib'
 - 'gh' the not code for 'well'. So, statement I is sufficient.
 - **Statement II:**
 - ✓ 'time is crucial' is coded as gh, ty, pd' ----(4)
 - ✓ 'trip went well' is coded as 'ib, nj, dr' ----(5)
 - From (1) and (5)
 - ✓ 'well' is coded as 'ib'
 - 'gh' the not code for 'well'. So, statement II is sufficient.
 - Data in either Statement I or Statement II is sufficient to answer the question.
- Hence, option C.

Q 143. Ans. B

Explanation:

Using the above information, we get the following arrangement:



We get H sitting third to the left of G.
Hence, option B is correct.

Q 144. Ans. C

Explanation:

- **Checking C1:** $O > F$
 - ✓ Here, $O < K \leq J = F$, we get the common sign between O and F is '<', but the given conclusion is $O > F$ so, it is not true.
- **Checking C2:** $G < K$
 - ✓ Here, $G < F = J \geq K$, we get opposite signs between G and K, so the given conclusion $G < K$ is not true.
- Hence, option C is correct.

Q 145. Ans. A

Explanation:

- Providing additional training and support to the struggling team members will help them improve

their performance and meet their targets. This approach addresses the issue at its root by enhancing the skills and capabilities of the team.

- Hence, option A is correct.

Q 146. Ans. C

Explanation:

- According to the question,
 - ✓ Principal (P) = ₹5000
 - ✓ Rate (R) = R%
 - ✓ Time (T) = R/2 years
 - ✓ Simple Interest (SI) = ₹10000
- Formula for simple interest,
 - ✓ $SI = (P \times R \times T) / 100$
 - ✓ $10000 = (5000 \times R \times R/2) / 100$
 - ✓ $10000 = 25R^2$
 - ✓ $R^2 = 400$
 - ✓ $R = 20$

Q 147. Ans. B

Explanation:

- According to the question,
- The ages of the five family members are:
 - ✓ 8 years, 14 years, 22 years, 36 years, and 45 years
 - ✓ Total age = $8 + 14 + 22 + 36 + 45 = 125$ years
 - ✓ Number of members = 5
 - ✓ Average age = $125 / 5 = 25$ years

Q 148. Ans. B

Explanation:

- According to the question,
 - ✓ Rahul : Rohit = 7 : 9
- Let their present ages be 7x years and 9x years,

respectively.

- ✓ Rohit is 10 years older than Rahul.
- ✓ $9x - 7x = 10$
- ✓ $2x = 10$
- ✓ $x = 5$

- Therefore,

- ✓ Rahul's present age = $7 \times 5 = 35$ years
- ✓ Rohit's present age = $9 \times 5 = 45$ years
- ✓ Rohit's age five years ago = $45 - 5 = 40$ years

Q 149. Ans. D

Explanation:

According to the question,

$$\sqrt[3]{343} = 7$$

$$\sqrt{64} = 8$$

$$\sqrt[3]{1000} = 10$$

$$\sqrt{196} = 14$$

$$\text{LCM of } (7, 8, 10, 14) = 280$$

Q 150. Ans. A

Explanation:

According to the question, the sides of the triangle are 18 cm, 24 cm, and 30 cm.

$$\text{Since } 18^2 + 24^2 = 30^2,$$

It is a right-angled triangle.

$$\text{Area of the large triangle} = \frac{1}{2} \times 18 \times 24 = 216 \text{ cm}^2$$

The triangle formed by joining the midpoints of the sides (Medial Triangle) is similar to the original triangle, and its area is $\frac{1}{4}$ of the area of the original triangle.

Therefore, the area of the smaller triangle =

$$\frac{1}{4} \times 216 = 54 \text{ cm}^2$$

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